

Can co-folding confidence tell interacting IDR pairs apart?

Stage 1 of the IDR-IDR benchmark: 60 of 62 AlphaFold Server jobs scored, five samples each. The pre-registered primary metric is ipTM of the top-ranked model and the pre-registered comparator is a leave-one-out logistic model on sequence composition alone - not chance. A test passes only if the paired-bootstrap CI on delta-AUC excludes zero.

THE ANSWER, ON THESE 60 JOBS

No - ipTM does not beat sequence composition.

Every pre-registered test fails its own criterion, and in the homotypic arm AF3 is significantly worse than the trivial baseline.

JOBS SCORED 60 / 62 S20, S21 pending	TEST 1 DELTA AUC -0.156 [-0.32, +0.01] FAIL	TEST 2B "THE RESULT" -0.107 [-0.26, +0.05] FAIL	HOMOTYPIC ARM -0.337 CI excludes zero: worse	GCN4 CANARY P21 0.73 pipeline behaves
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The pre-registered tests *primary metric ipTM; comparator = composition baseline*

test / group	n pos/neg	AF3 AUC	95% CI	baseline	delta	delta 95% CI	verdict
Test 1 17 positives vs 31 real negatives	17/31	0.611	[0.45, 0.77]	0.767	-0.156	[-0.32, +0.01]	FAIL
Test 2a mutual folding vs hard zipper	4/3	0.417	[0.00, 0.88]	0.250	+0.167	[-0.50, +0.88]	FAIL
Test 2b fuzzy positives (the result)	17/28	0.676	[0.51, 0.83]	0.784	-0.107	[-0.26, +0.05]	FAIL
Test 3 vs gold-standard negatives	17/8	0.118	[0.00, 0.30]	0.640	-0.522	[-0.74, -0.29]	FAIL
Test 4 vs scrambles (separate arm)	17/8	0.566	[0.29, 0.83]	--	--	--	
Homotypic arm	12/11	0.564	[0.30, 0.83]	0.902	-0.337	[-0.58, -0.12]	FAIL
Heterotypic arm	5/20	0.485	[0.23, 0.74]	0.500	-0.015	[-0.32, +0.29]	FAIL

Test 4 has no baseline column by design: scrambles are composition-matched, so the composition model cannot separate them from their parents and is not a meaningful comparator in that arm.

READS AS The plan's own decision table says: delta-AUC_fuzzy near zero with AUC_fold high is the clean expected split; both low means debug first. Here delta-AUC_fuzzy is -0.11 with a CI crossing zero, and AUC_fold is 0.42 - not high. Pages 3 to 4 show why that is not a broken pipeline but a real failure mode: AF3 dimerises any zipper-competent pair, cognate or not.

Does AF3 beat composition? Every grouping.

Delta-AUC = AUC(AF3 ipTM) - AUC(composition baseline), paired bootstrap, 5,000 resamples. Right of the zero line, AF3 adds information over composition; left, it subtracts. A group passes only if the whole interval sits right of zero.

Pre-registered

Test 1 17 positives vs 31 real negatives (17/31)

Test 2a mutual folding vs hard zipper (4/3)

Test 2b fuzzy positives (the result) (17/28)

Test 3 vs gold-standard negatives (17/8)

By pair type

Homotypic arm (12/11)

Heterotypic arm (5/20)

By negative type

vs noncognate (17/10)

vs cross_kingdom (17/5)

vs charge_repulsive (17/2)

vs hard_coiledcoil (17/3)

vs homo_negative (17/11)

By negative evidence

vs demonstrated (17/5)

vs demonstrated+matched_internal (17/3)

vs gold (both demonstrated grades) (17/8)

vs inferred (17/11)

vs absence_only (17/10)

vs uncertain (17/2)

vs weak (inferred + absence_only) (17/21)

By positive subgroup

Fuzzy heterotypic positives (5/31)

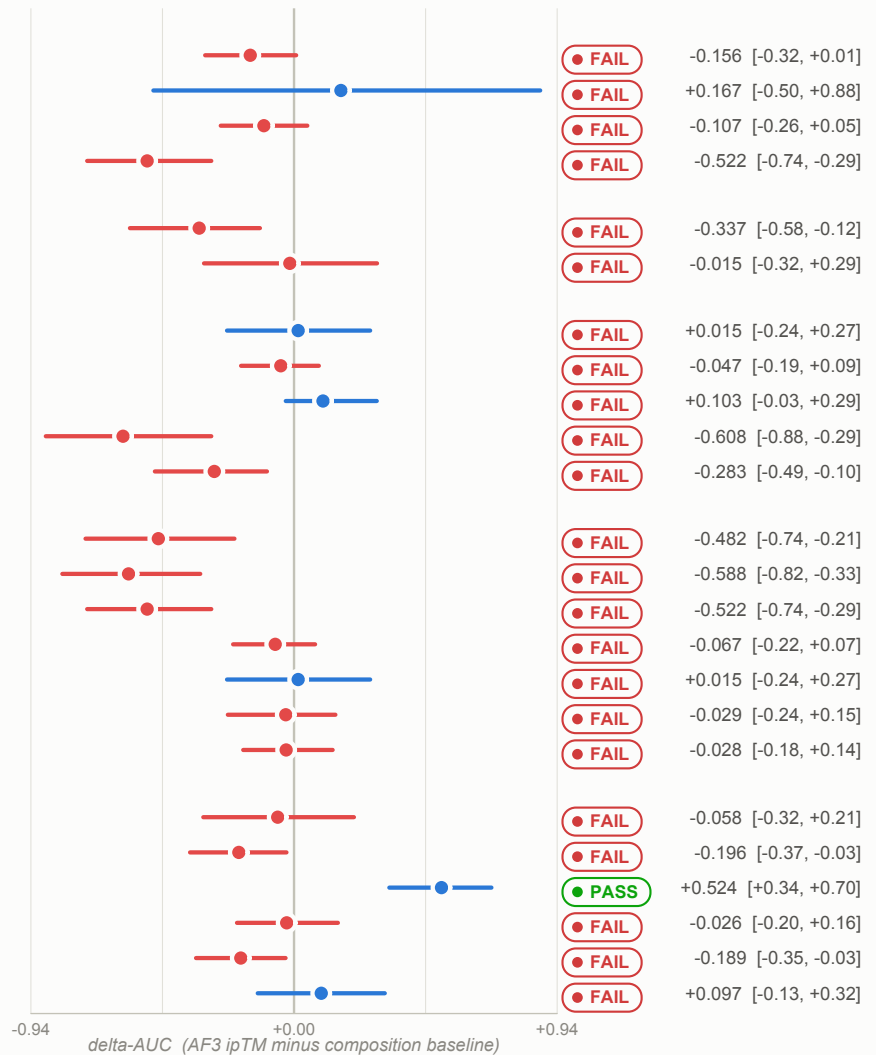
Fuzzy homotypic positives (12/31)

Mutual-folding (memorised) positives (4/31)

All 21 positives incl. memorised (21/31)

LLPS-reported positives (15/31)

Positives without LLPS report (2/31)



READS AS

Exactly one grouping clears zero on ipTM, and it is the pipeline control: the four memorised complexes, at +0.52. Of the other 23, two exclude zero on the wrong side - the homotypic arm (-0.34) and the gold-standard negatives (-0.52), where AF3 ranks the negatives above the positives - and the rest are indistinguishable from composition.

Is the pipeline sound?

Before reading a null result, the controls have to hold. They do: the canary folds, the memorised complexes score highest in the set, and every job ran the sequence it was supposed to.

Memorised controls and the canary

pipeline controls, excluded from every headline number

control	PDB	residues	ipTM	mean pLDDT	interchain PAE	ranking	sample sd
P21 GCN4_LZ x GCN4_LZ	2ZTA	66	0.73	88.1	4.1	1.24	0.027
P20 MAX_bHLHZ x MAX_bHLHZ	1AN2	162	0.80	86.8	6.1	0.95	0.005
P07 FOS_bZIP x JUN_bZIP	1FOS	129	0.74	86.6	8.5	1.03	0.005
P06 NCOA3_ACTR x CBP_NCBD	1KBH	106	0.70	76.9	6.3	0.89	0.021

GCN4-p1 (P21, 66 residues) is the pre-registered canary: if AF3 could not dimerise it confidently the instruction was to stop and debug. It reaches ipTM 0.73 with mean pLDDT 88 and a 4.1 Å interchain PAE, on an MSA of only 23 paired sequences. All four memorised complexes land between 0.70 and 0.80, the top of the whole set.

Data integrity

checked against the job requests the server returned

check	result	
Submitted sequences match the pair table	all 60 jobs, both chains exact	● PASS
Samples per job	5 of 5 for every job (300 summary files, 300 full-data files)	● PASS
Job-to-row matching	60 matched by name, 0 ambiguous	● PASS
Model seeds	59 jobs on seed 1; N24 on a server-assigned seed	● WARN
Steric clashes	a clash in some sample of P13 and P17 only (2 of 60)	● WARN
Structure templates	on for all 95 chain entries - server default, not a choice	● WARN
Jobs outstanding	S20 and S21 (batch 4) not yet run	● NOTE

Sample spread

five samples from one seed, so this is sample spread, not seed replication

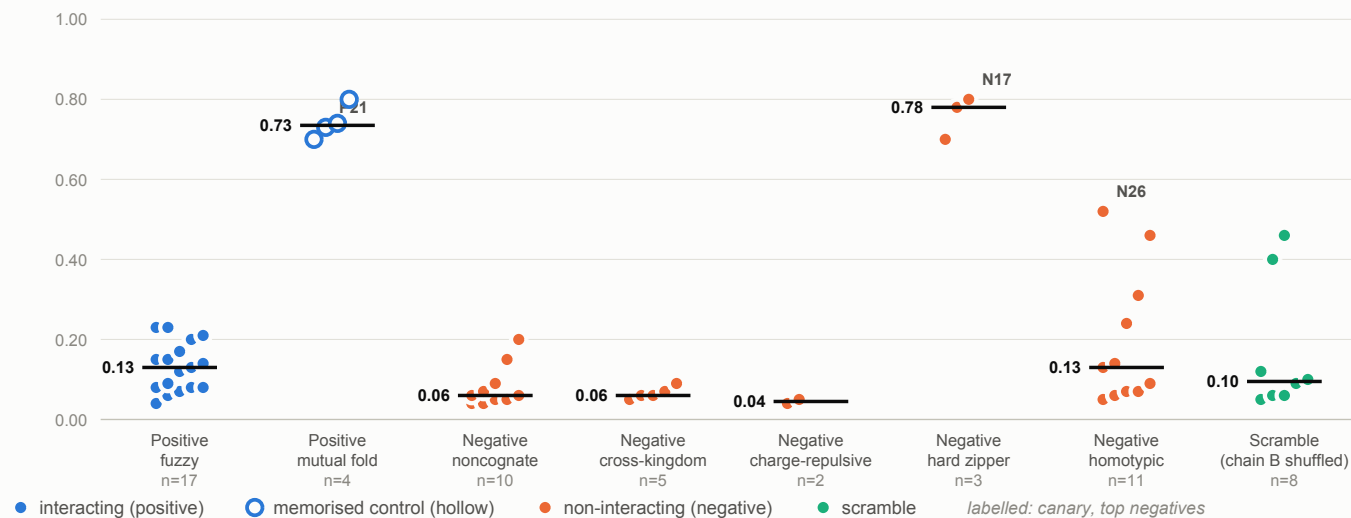
Fuzzy positives: mean within-job ipTM sd 0.0141 (n=17). Mutual-folding positives: 0.0146 (n=4). The plan expected fuzzy pairs to be the more variable; they are not, because all five samples come from one seed and share a trunk embedding. Seed replicates are still needed to measure run-to-run variance.

READS AS

Nothing here explains the null result away. The controls behave, so the failures on pages 1 and 2 are about AF3's discrimination, not about the harness - with one caveat that cuts the other way: templates were on for every chain, which can only help the controls look good.

Where every job actually lands

One dot per job, ipTM of the top-ranked sample. Black tick is the class median. The dots are the whole dataset: no aggregation, nothing hidden.



READS AS The three hard-zipper negatives (N15, N16, N17) sit at 0.70-0.80 - level with the memorised true complexes - while the 17 genuine fuzzy positives have a median ipTM of 0.13 (range 0.04-0.23). AF3 confidence tracks "can these two helices form a coiled coil", not "do these two regions interact".

Primary metric, full statistics *permutation p, BH-corrected across the 16-metric panel*

group	n	ipTM AUC	95% CI	perm p	BH q	composition	length alone
Test 1 17 positives vs 31 real negatives	17/31	0.611	[0.45, 0.77]	0.2069	0.2758	0.767	0.676
Test 2a mutual folding vs hard zipper	4/3	0.417	[0.00, 0.88]	0.8601	1.0000	0.250	0.417
Test 2b fuzzy positives (the result)	17/28	0.676	[0.51, 0.83]	0.0467	0.0721	0.784	0.641
Test 3 vs gold-standard negatives	17/8	0.118	[0.00, 0.30]	0.0010	0.0035	0.640	0.985
Test 4 vs scrambles (separate arm)	17/8	0.566	[0.29, 0.83]	0.6202	0.7634	--	0.684
Homotypic arm	12/11	0.564	[0.30, 0.83]	0.6081	0.7271	0.902	0.795
Heterotypic arm	5/20	0.485	[0.23, 0.74]	0.9390	1.0000	0.500	0.530

The last two columns are the reference predictors on the same jobs: a composition-only model, and total chain length on its own. Where they beat the ipTM column, AF3 has added nothing a sequence length could not.

Homotypic and heterotypic, analysed apart

Plan criterion 3. The two arms have different baselines and different MSA situations, so pooling them lets chain identity leak into the answer. Each arm is scored against a baseline refitted inside that arm.

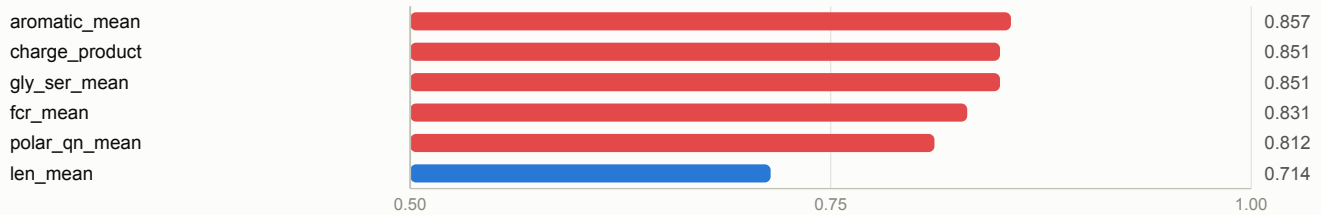
arm	n	ipTM AUC	95% CI	arm baseline	delta	delta 95% CI	verdict	contact sum AUC
Homotypic arm	12/11	0.564	[0.30, 0.83]	0.902	-0.337	[-0.58, -0.12]	● FAIL	0.970
Heterotypic arm	5/20	0.485	[0.23, 0.74]	0.500	-0.015	[-0.32, +0.29]	● FAIL	0.465

READS AS The homotypic arm is where the benchmark is decided, and it is where AF3 loses outright: ipTM 0.56 against a composition baseline of 0.90, delta -0.34 with the interval clear of zero. In the heterotypic arm composition is useless (0.50) and AF3 is no better (0.49) - on only 5 positives.

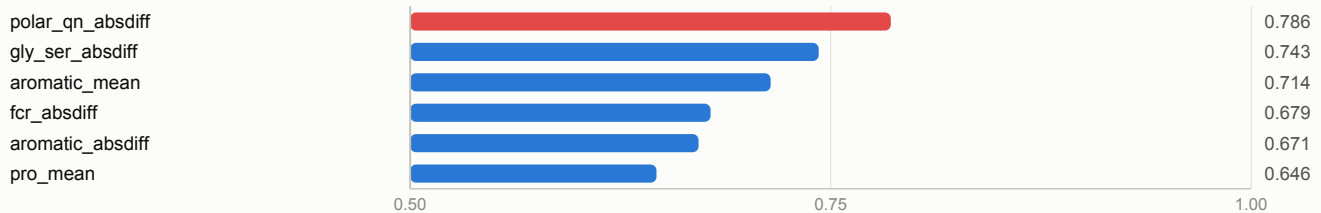
Why composition is so strong in the homotypic arm

single-feature AUC, two-sided, from baseline_leakage.py

Homotypic pairs (14 positive / 11 negative)



Heterotypic pairs (7 positive / 20 negative)



A single feature above 0.75, or a leave-one-out model above 0.80, is the documented leakage threshold. In the homotypic arm five features clear it on their own: LC domains that self-associate really are aromatic-rich and weakly charged. That is biology, not a curation error - but it means a homotypic claim needs to beat 0.85, not 0.5.

By type of negative

The plan asks for negative subtypes separately, because they are not equally clean: non-cognate human pairs are the fair test, cross-kingdom pairs share the scramble arm's missing-MSA problem in milder form, and the hard zippers are adversarial by design.

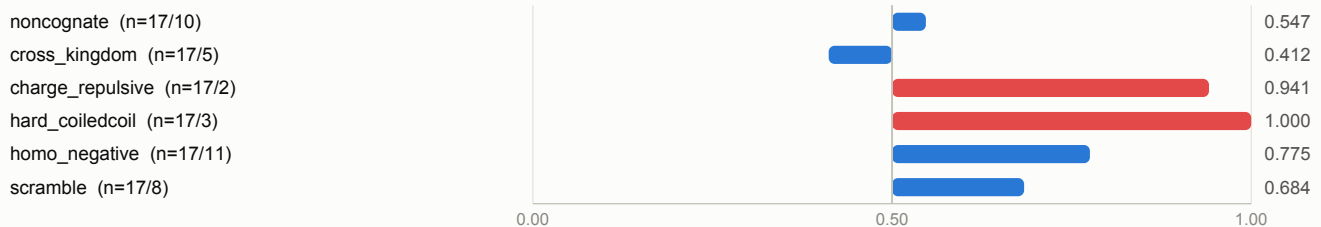
group	n	ipTM AUC	95% CI	BH q	compos.	delta	delta CI	length	contact s..
noncognate	17/10	0.768	[0.56, 0.94]	0.0291	0.753	+0.015	[-0.24, +0.27]	0.547	0.829
cross_kingdom	17/5	0.847	[0.66, 0.99]	0.0393	0.894	-0.047	[-0.19, +0.09]	0.412	0.900
charge_repulsive	17/2	0.956	[0.82, 1.00]	n<3	0.853	+0.103	[-0.03, +0.29]	0.941	0.971
hard_coiledcoil	17/3	0.000	[0.00, 0.00]	0.0034	0.608	-0.608	[-0.88, -0.29]	1.000	0.706
homo_negative	17/11	0.465	[0.22, 0.71]	0.8844	0.749	-0.283	[-0.49, -0.10]	0.775	0.684
scramble	17/8	0.566	[0.29, 0.83]	0.7634	--	--	--	0.684	0.783

Charge-repulsive has only 2 negatives, so no permutation test is reported for it. Scrambles have no composition comparator by construction.

READS AS The subtype decides the answer. AF3 scores highest against the negatives whose second chain has the weakest evolutionary signal - charge-repulsive (0.96, only 2 pairs) and cross-kingdom (0.85) - holds up against non-cognate human pairs (0.77), then inverts completely against the hard zippers (0.00, where length alone scores a perfect 1.00 because every zipper pair is short) and against the homotypic negatives (0.47). The cleanest negatives and the nastiest ones disagree, which is the whole reason the subtypes are reported apart.

The same groups, by what the model could be reading instead

Total chain length alone, as a classifier (0.5 = chance)



Bars to the right of the centre line mean the positives are longer; to the left, the negatives are. Two of these groups are separated almost perfectly by length before AF3 is consulted at all.

By how well we know the negative is negative

Every negative carries an evidence grade. 21 of 31 rest on inferred or absence-only evidence; 8 are demonstrated. The plan says that if the headline and the gold-standard subset disagree, believe the gold standard.

group	n	ipTM AUC	95% CI	BH q	compos.	delta	delta CI	length	contact s..
demonstrated	17/5	0.188	[0.00, 0.45]	0.0770	0.671	-0.482	[-0.74, -0.21]	0.976	0.682
demonstrated+matched_internal	17/3	0.000	[0.00, 0.00]	0.0084	0.588	-0.588	[-0.82, -0.33]	1.000	0.706
gold (both demonstrated grades)	17/8	0.118	[0.00, 0.30]	0.0035	0.640	-0.522	[-0.74, -0.29]	0.985	0.691
inferred	17/11	0.805	[0.59, 0.98]	0.0182	0.872	-0.067	[-0.22, +0.07]	0.647	0.832
absence_only	17/10	0.768	[0.56, 0.94]	0.0291	0.753	+0.015	[-0.24, +0.27]	0.547	0.829
uncertain	17/2	0.735	[0.47, 0.94]	n<3	0.765	-0.029	[-0.24, +0.15]	0.235	0.706
weak (inferred + absence_only)	17/21	0.787	[0.62, 0.93]	0.0032	0.815	-0.028	[-0.18, +0.14]	0.599	0.831

Uncertain has 2 rows (H1.0 alone, ERD10 alone) and is descriptive only.

READS AS They disagree violently - and the gold standard is the confounded one. Against the 8 demonstrated negatives ipTM inverts to 0.12, but those 8 negatives are 94-236 residues while the 17 positives are 205-882, so length alone scores 0.985 on that exact split. The inversion is a length and foldability artefact, not evidence that AF3 ranks real complexes below non-complexes.

What the gold-standard subset is made of *the 8 demonstrated negatives, ranked by ipTM*

gold-standard negative	subtype	evidence	residues	ipTM	PAE	pLDDT	contact s..
N17 GCN4_LZ x JUN_bZIP	hard_coiledcoil	dem.	100	0.80	5.9	88	47
N15 FOS_bZIP x MAX_bHLHZ	hard_coiledcoil	dem.	143	0.78	6.1	88	73
N16 JUN_bZIP x MAX_bHLHZ	hard_coiledcoil	dem.	148	0.70	9.4	87	72
N26 NCOA3_ACTR x NCOA3_ACTR	homo_negative	dem.+matched_in..	94	0.52	8.5	70	62
N27 CBP_NCBD x CBP_NCBD	homo_negative	dem.+matched_in..	118	0.46	12.5	67	52
N28 FOS_bZIP x FOS_bZIP	homo_negative	dem.+matched_in..	124	0.31	20.7	74	61
N31 EIF4EBP1 x EIF4EBP1	homo_negative	dem.	236	0.14	27.1	48	111
N29 P53_TAD x P53_TAD	homo_negative	dem.	122	0.13	22.2	55	64

Six of the eight are short helix- or zipper-competent constructs. Three of them (N15, N16, N17) are exactly the adversarial coiled-coil negatives, and three more (N26, N27, N28) are single chains of a real complex run against themselves.

By type of positive

All rows below are scored against the same 31 real-sequence negatives, so the groups are directly comparable.

group	n	ipTM AUC	95% CI	BH q	compos.	delta	delta CI	length	contact s..
Fuzzy heterotypic positives	5/31	0.394	[0.19, 0.61]	0.6958	0.452	-0.058	[-0.32, +0.21]	0.600	0.300
Fuzzy homotypic positives	12/31	0.702	[0.54, 0.85]	0.0509	0.898	-0.196	[-0.37, -0.03]	0.707	0.989
Mutual-folding (memorised) posi..	4/31	0.944	[0.86, 1.00]	0.0017	0.419	+0.524	[+0.34, +0.70]	0.129	0.847
All 21 positives incl. memorised	21/31	0.674	[0.52, 0.81]	0.0642	0.700	-0.026	[-0.20, +0.16]	0.571	0.798
LLPS-reported positives	15/31	0.604	[0.44, 0.77]	0.3170	0.794	-0.189	[-0.35, -0.03]	0.690	0.801
Positives without LLPS report	2/31	0.661	[0.45, 0.87]	n<3	0.565	+0.097	[-0.13, +0.32]	0.565	0.677

The LLPS rows are a stratification, never an outcome: a 1:1 co-fold cannot report on phase separation, which is many-body and concentration-dependent.

READS AS

The four memorised complexes are the positives AF3 separates cleanly (ipTM AUC 0.94); fuzzy positives sit at 0.39 heterotypic and 0.70 homotypic, and reported LLPS makes no difference (0.60). Memorisation, not disorder, is what ipTM tracks here. One exception is worth flagging honestly: for the 12 fuzzy homotypic positives, contact probability sum beats composition by +0.091 [+0.011, +0.204] - the only cell outside the memorised controls to clear zero in all 464. It is exploratory, uncorrected for the 464 comparisons, and the pre-registered homotypic arm (which uses homotypic negatives and the arm baseline) does not pass. Treat it as the hypothesis to test on a fresh set, not a result.

The sharpest tests in the set

matched internal contrasts - same chains, same MSAs, only the pairing differs

These three comparisons do not rest on a literature absence at all. ACTR and NCBD each fold only with the other, and c-Fos forms no stable homodimer, so ACTR x ACTR, NCBD x NCBD and Fos x Fos are negatives whose chains also appear in a positive. The plan calls the outcome unambiguous either way.

job	label	residues	ipTM	interchain PAE	mean pLDDT	contact sum
P06 NCOA3_ACTR x CBP_NCBD - ACTR and NCBD each fold only with the other						
P06 NCOA3_ACTR x CBP_NCBD	positive	106	0.70	6.3	76.9	66
N26 NCOA3_ACTR x NCOA3_ACTR	negative	94	0.52	8.5	70.3	62
N27 CBP_NCBD x CBP_NCBD	negative	118	0.46	12.5	67.3	52
P07 FOS_bZIP x JUN_bZIP - c-Fos forms no stable homodimer (O'Shea 1989)						
P07 FOS_bZIP x JUN_bZIP	positive	129	0.74	8.5	86.6	66
N28 FOS_bZIP x FOS_bZIP	negative	124	0.31	20.7	74.4	61

READS AS

AF3 gets all three orderings right on ipTM, and on 12 of 16 metrics for P06 and 15 of 16 for P07. This is the one genuinely encouraging result in stage 1 - with the caveat on page 11 that both positives were handed per-chain templates taken from their own cognate complex.

The whole 16-metric panel

ipTM may simply be floored for disordered pairs, so the plan pre-registered a 16-metric panel with permutation p-values and Benjamini-Hochberg correction. Every metric is oriented so that a higher AUC means "more interaction".

AUC by metric and group

A dot in the corner marks BH $q < 0.05$ within that group. Cell values are AUCs; .50 is chance.

	ipTM	chain-pair ipTM	ranking score	PTM	fraction disordered	chain-pair PAE min	interchain PAE mean	interchain PAE p05	PAE frac < 10 A	PAE frac < 5 A	contact prob. sum	contact prob. density	contact prob. max	contacts > 0.5	contact particip. ratio	mean pLDDT
Test 1 (17/31)	.61	.61	.57	.47	.39	.68	.56	.62	.62	.60	.79 [*]	.61	.68	.67	.70	.29
Test 2a (4/3)	.42	.42	.54	.42	.50	.62	.50	.58	.58	.67	.42	.75	.50	.46	.67	.17
Test 2b (17/28)	.68	.68	.63	.52	.41	.75 [*]	.62	.69	.68 [*]	.67 [*]	.80 [*]	.66	.75 [*]	.72 [*]	.68	.32
Test 3 (17/8)	.12 [*]	.12 [*]	.24	.07 [*]	.22 [*]	.30	.01 [*]	.11 [*]	.26	.28	.69	.12 [*]	.26	.43	.69	.12 [*]
Test 4 (17/8)	.57	.57	.62	.51	.31	.71	.49	.58	.60	.64	.78	.54	.73	.68	.61	.60
Homotypic arm (12/11)	.56	.56	.62	.36	.30	.80	.44	.59	.68	.64	.97 [*]	.53	.80	.79	.55	.23
Heterotypic arm (5/20)	.48	.48	.35	.57	.55	.42	.45	.46	.38	.42	.47	.46	.48	.42	.58	.50

AUC, diverging around chance (0.50)
 0.15 inverted discriminating 0.85

Delta-AUC against the composition baseline, same cells

Blank row for test 4: scrambles have no composition comparator.

	ipTM	chain-pair ipTM	ranking score	PTM	fraction disordered	chain-pair PAE min	interchain PAE mean	interchain PAE p05	PAE frac < 10 A	PAE frac < 5 A	contact prob. sum	contact prob. density	contact prob. max	contacts > 0.5	contact particip. ratio	mean pLDDT
Test 1 (17/31)	-.16	-.16	-.20	-.29	-.38	-.09	-.21	-.15	-.15	-.16	.02	-.16	-.09	-.09	-.07	-.47
Test 2a (4/3)	.17	.17	.29	.17	.25	.38	.25	.33	.33	.42	.17	.50	.25	.21	.42	-.08
Test 2b (17/28)	-.11	-.11	-.16	-.26	-.38	-.03	-.17	-.10	-.10	-.12	.01	-.13	-.03	-.06	-.11	-.46
Test 3 (17/8)	-.52	-.52	-.40	-.57	-.42	-.34	-.62	-.53	-.37	-.36	.05	-.51	-.37	-.21	.05	-.52
Test 4 (17/8)																
Homotypic arm (12/11)	-.34	-.34	-.28	-.54	-.61	-.11	-.46	-.31	-.22	-.26	.07	-.37	-.11	-.11	-.35	-.67
Heterotypic arm (5/20)	-.02	-.02	-.15	.07	.05	-.08	-.05	-.04	-.12	-.08	-.03	-.04	-.02	-.08	.08	.00

delta-AUC
 -0.30 worse than composition better +0.30

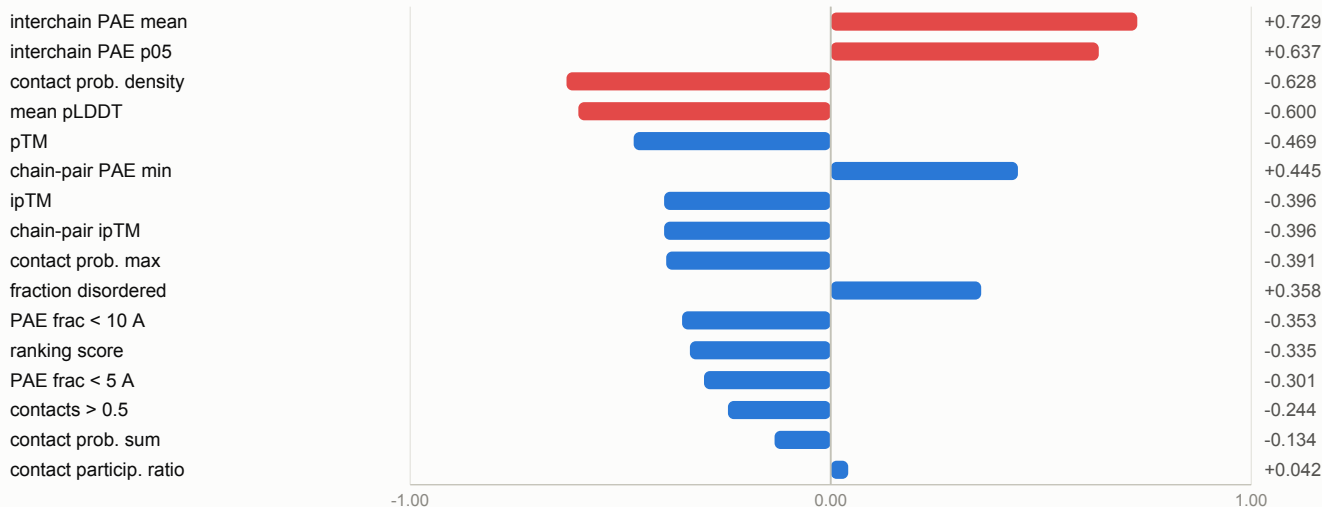
READS AS

Read the two grids together. Several metrics beat chance in the top grid - contact probability sum reaches 0.79 at $q = 0.008$, and 0.97 in the homotypic arm - yet the bottom grid is essentially all grey or red: across these seven groups not one metric beats composition with a CI clear of zero. The best single cell is contact prob. density at delta +0.500 (test 2a, $n=4/3$), and its interval still crosses zero. The one non-control exception anywhere in the 464 cells is on page 8.

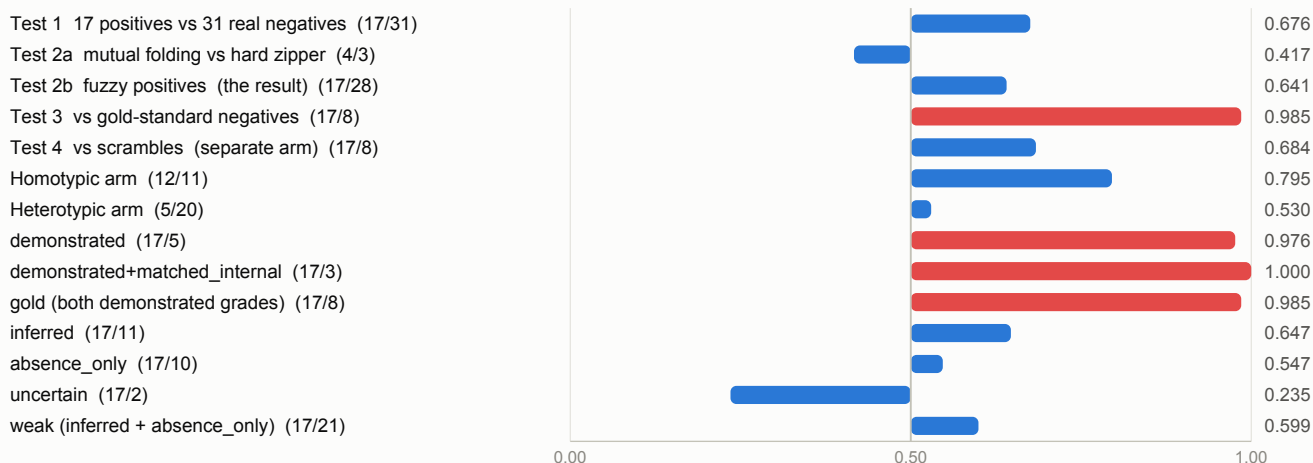
Confound 1: chain length

The plan flags any metric whose Spearman rho against total residue count exceeds 0.5, and requires it to be reported partialled as well as raw. Four metrics are flagged - and they are exactly the ones that carry the gold-standard inversion.

Spearman rho between metric and total residue count, all 60 jobs

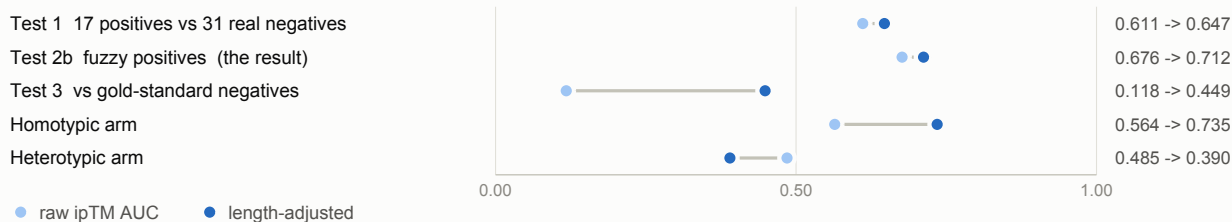


Length alone, as a classifier, in every group



Raw versus length-adjusted

metrics residualised on log total residues, then rescored



READS AS

Adjusting for length moves the gold-standard test from 0.12 to 0.45 and barely touches the headline test. So the dramatic inversion was length; the null result was not. This is a sensitivity analysis, not a pre-registered test.

Confound 2: what the model was given

Two inputs differ systematically between the arms of this benchmark, and neither was a design decision.

Structure templates were on for every chain server default; the batch files never set it

The uploaded batch files are dialect version 1 and contain no template switch. The server upgraded them to version 3 and set `useStructureTemplate = true` on all 95 chain entries, then returned 2 to 4 template hits per chain for all 60 jobs. For the matched internal negatives that matters directly: 6es7 - a structure containing both ACTR and NCBD, since it templates both chains of P06 - is also a template for N26 (ACTR x ACTR) and N27 (NCBD x NCBD), and 1fos, which templates both chains of P07, is a template in N15, N16, N17 and N28. AF3 applies templates within a chain, so this is the bound conformation of each partner rather than the interface itself - but it is exactly the help a zipper needs to dimerise.

job	label	template hits per chain (PDB entry)	unpaired MSA dept..	ipTM
P06 NCOA3_ACTR x CBP_NCBD	positive	a: 6es7, 6es5, 2c52, 3t47; b: 6es7, 6es6, 6sqc, 1jjs	a 379 b 774	0.70
N26 NCOA3_ACTR x NCOA3_ACTR	negative	a_b: 6es7, 6es5, 2c52, 3t47	a_b 379	0.52
N27 CBP_NCBD x CBP_NCBD	negative	b_a: 6es7, 6es6, 6sqc, 1jjs	b_a 774	0.46
P07 FOS_bZIP x JUN_bZIP	positive	a: 1fos, 5vpa, 1s9k, 1a02; b: 5vpa, 5t01, 1fos, 1t2k	a 1,478 b 1,259	0.74
N28 FOS_bZIP x FOS_bZIP	negative	b_a: 1fos, 5vpa, 1s9k, 1a02	b_a 1,478	0.31
N15 FOS_bZIP x MAX_bHLHZ	negative	a: 1fos, 5vpa, 1s9k, 1a02; b: 1r05, 1nlw, 1hlo, 5i4z	a 1,478 b 2,479	0.78
N17 GCN4_LZ x JUN_bZIP	negative	a: 5apu, 5apu, 5apw, 5apq; b: 5vpa, 5t01, 1fos, 1t2k	a 47 b 1,259	0.80
P21 GCN4_LZ x GCN4_LZ	positive	a_b: 5apu, 5apu, 5apw, 5apq	a_b 47	0.73

The scrambles really do lose their MSA confirming the plan's reason for keeping them a separate arm

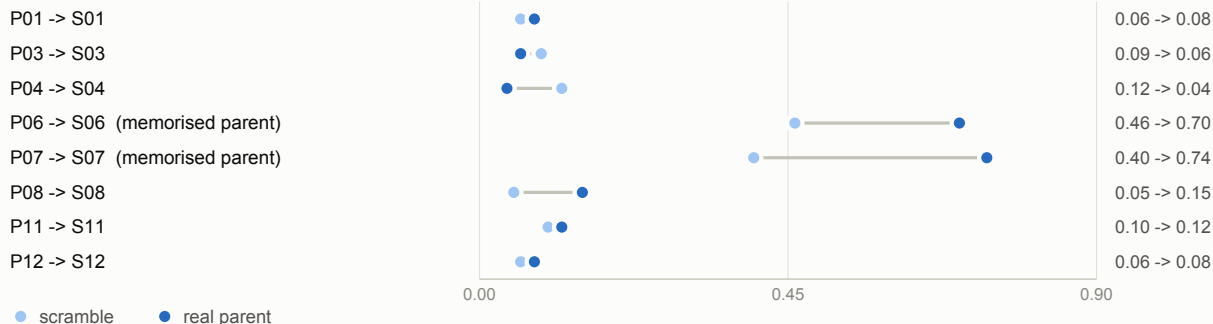
Unpaired MSA depth, shuffled chains: 4 to 34 sequences (median 8, n=8). Real chains: 47 to 14,534 (median 927, n=87). A shuffled sequence has no homologs, so low confidence there follows from missing evolutionary information rather than from a missing interface - which is why test 4 is reported apart and cannot carry the headline.

READS AS Both confounds push the same way: they make the memorised, foldable, short, template-rich pairs look good and the long fuzzy ones look bad. The single cheapest next experiment is to re-run a handful of jobs with `useStructureTemplate = false` and see how much of the control signal survives.

The scramble arm, paired

Each scramble shares chain A with its parent positive and differs only in a shuffled chain B, so the natural analysis is paired. Eight of the ten have run; S20 and S21 are pending.

ipTM: parent positive versus its composition-matched scramble



metric	parent beats scramble	sign-test p	median difference	unpaired AUC	BH q
ipTM	6 of 8	0.289	+0.020	0.566	0.7634
chain-pair ipTM	6 of 8	0.289	+0.020	0.566	0.7634
ranking score	6 of 8	0.289	+0.075	0.621	0.7397
pTM	6 of 8	0.289	+0.110	0.511	0.9792
fraction disordered	4 of 8	1.000	-0.010	0.309	0.3705
chain-pair PAE min	6 of 8	0.289	-5.910	0.713	0.3705
interchain PAE mean	6 of 8	0.289	-0.790	0.493	0.9792
interchain PAE p05	6 of 8	0.289	-4.050	0.581	0.7634
PAE frac < 10 A	4 of 8	1.000	+0.003	0.596	0.7580
PAE frac < 5 A	3 of 8	0.727	+0.000	0.643	0.5341
contact prob. sum	6 of 8	0.289	+9.985	0.783	0.3440
contact prob. density	6 of 8	0.289	+0.002	0.544	0.8736
contact prob. max	6 of 8	0.289	+0.315	0.728	0.3705
contacts > 0.5	4 of 8	1.000	+13.0	0.684	0.3705
contact particip. ratio	3 of 8	0.727	-248.0	0.614	0.7397
mean pLDDT	7 of 8	0.070	+7.822	0.596	0.7580

READS AS With 8 pairs the paired test has almost no power: ipTM favours the real parent in 6 of 8 (sign-test $p = 0.29$). There is no evidence either way yet on whether the little signal AF3 has depends on evolutionary information - finishing S20 and S21 makes it 10 pairs, which is still thin.

How every number here was produced

Nothing in this report is computed by hand. The statistics come from the project's own functions so the report cannot drift from the pipeline.

Data

60 of 62 AlphaFold Server jobs, five samples each, downloaded 14 September 2026. Per job the reported value is the sample with the highest ranking_score, matching parse_af3_results.py. S20 and S21 (scrambles of the two memorised homodimers) are not yet run.

AUC

Mann-Whitney AUC with tie correction, on the metric oriented so higher means more interaction. 0.5 is chance; below 0.5 means the negatives outrank the positives.

Confidence intervals

Percentile bootstrap, 10,000 resamples, resampling positives and negatives independently.

p and q

Two-sided permutation test on the label assignment, 10,000 permutations, then Benjamini-Hochberg across the 16-metric panel within each group. Groups with fewer than 3 on either side are reported as descriptive only.

Composition baseline

Leave-one-out logistic regression on 24 cheap pair features (NCPR, FCR, aromatic and G/S content, hydropathy, length, charge complementarity, homotypic flag - means and absolute differences per pair). Fitted on the 52 real-sequence jobs for pooled groups, and refitted inside each arm for the homotypic and heterotypic rows. Raw LOO AUC: pooled 0.700, homotypic 0.851, heterotypic 0.471 (0.529 two-sided).

Delta-AUC

Paired bootstrap on AUC(AF3) - AUC(baseline), 5,000 resamples, resampling the same jobs for both score vectors. A test passes only if the 95% interval excludes zero. This is the pre-registered primary criterion; beating 0.5 is not the bar.

Pre-registered versus exploratory

Pre-registered: the five subsets on page 1, the homotypic/heterotypic split, the 16-metric panel with BH correction, and the length confound flag. Exploratory, and labelled as such: the negative-subtype, evidence-grade and positive-subgroup groupings, the length-adjusted sensitivity analysis, and the paired scramble test.

Known caveats carried from the plan

Structure templates were on for all 95 chain entries (server default). 59 jobs ran on seed 1 and N24 on a server-assigned seed. Five samples share one seed, so spread is sample spread, not seed replication. 21 of 31 negatives rest on inferred or absence-only evidence. Length is not matched by design. Two negatives (H1.0 alone, ERD10 alone) are graded uncertain and should be confirmed or dropped. has_clash appears in the plan's metric list but not in the parser's panel; it is true in some sample of P13 and P17 only.

Files

af3_report_data.json holds every number in this report; af3_all_statistics.csv is the full 496-cell table (31 groups x 16 metrics); af3_scored.csv is the per-job scored table written by parse_af3_results.py.

Appendix A - every job, every headline metric

All 60 scored jobs. This is the table view for every figure in the report: each plotted dot appears here as a number.

ID	chains	label	class	res	ipTM	sd	pTM	PAE	pLDDT	C sum	C>.5	disor..	comp..
N01	FUS_LC x PTMA_full	negative	noncognate	274	0.04	0.000	0.09	31.6	47.4	0	0	1.00	-4.67
N02	HNRNPA1_LC x H10_full	negative	noncognate	329	0.06	0.004	0.27	31.2	56.5	4	0	0.79	-4.25
N03	TDP43_CTD x NPM1_acidic	negative	noncognate	269	0.05	0.000	0.13	31.0	49.1	0	0	0.25	-3.57
N04	EWSR1_LC x SNCA_full	negative	noncognate	405	0.05	0.004	0.12	31.6	53.5	0	0	1.00	-0.17
N05	TAF15_LC x MAX_bHLHZ	negative	noncognate	289	0.20	0.029	0.29	29.1	55.0	35	0	0.81	-1.73
N06	DDX4_NIDR x H10_full	negative	noncognate	430	0.07	0.004	0.23	31.5	48.3	1	0	0.84	+0.85
N07	CAPRIN1_CT x GCN4_LZ	negative	cross_kingdom	136	0.09	0.038	0.24	28.4	68.2	26	0	1.00	-0.57
N08	G3BP1_IDR1 x NPM1_acidic	negative	charge_repulsive	200	0.04	0.000	0.10	30.4	56.6	0	0	0.38	-2.24
N09	PTMA_full x NPM1_acidic	negative	charge_repulsive	232	0.05	0.005	0.10	30.8	50.4	0	0	0.54	-1.12
N10	FUS_LC x ERD10_full	negative	cross_kingdom	423	0.06	0.000	0.12	31.6	52.3	0	0	1.00	-3.37
N11	TDP43_CTD x ERD10_full	negative	cross_kingdom	408	0.07	0.004	0.12	31.4	51.8	0	0	1.00	-2.94
N12	NSP1_FG x H10_full	negative	noncognate	369	0.09	0.009	0.27	30.4	45.8	0	0	0.56	-3.13
N13	SRSF1_RS x PTMA_full	negative	noncognate	163	0.06	0.007	0.10	27.9	51.4	12	0	0.50	+3.73
N14	U170K_RS x NPM1_acidic	negative	noncognate	328	0.15	0.016	0.29	29.6	73.2	11	0	1.00	+0.07
N15	FOS_bZIP x MAX_bHLHZ	negative	hard_coiledcoil	143	0.78	0.008	0.78	6.1	88.5	73	70	0.38	-2.18
N16	JUN_bZIP x MAX_bHLHZ	negative	hard_coiledcoil	148	0.70	0.008	0.75	9.4	87.2	72	70	0.36	+0.85
N17	GCN4_LZ x JUN_bZIP	negative	hard_coiledcoil	100	0.80	0.007	0.74	5.9	87.5	47	45	0.76	+1.82
N18	HNRNPA2_LC x ERD10_full	negative	cross_kingdom	420	0.06	0.000	0.12	31.5	52.3	0	0	1.00	-4.67
N19	UBQLN2_CT x SNCA_full	negative	noncognate	315	0.04	0.004	0.18	31.6	62.7	16	0	0.30	+0.54
N20	CAPRIN1_CT x ERD10_full	negative	cross_kingdom	363	0.05	0.000	0.11	31.5	57.3	0	0	1.00	-1.08
N21	PTMA_full x PTMA_full	negative	homo_negative	222	0.06	0.006	0.13	30.8	53.2	37	0	1.00	-4.08
N22	H10_full x H10_full	negative	homo_negative	388	0.07	0.005	0.24	30.9	52.4	32	0	0.63	-1.38
N23	NPM1_acidic x NPM1_acidic	negative	homo_negative	242	0.07	0.005	0.10	29.9	51.5	39	0	0.96	-1.74
N24	ERD10_full x ERD10_full	negative	homo_negative	520	0.09	0.005	0.12	31.2	44.1	44	0	1.00	-0.44
N25	G3BP1_IDR1 x G3BP1_IDR1	negative	homo_negative	158	0.05	0.005	0.10	29.2	52.8	34	0	1.00	-5.43
N26	NCOA3_ACTR x NCOA3_ACTR	negative	homo_negative	94	0.52	0.136	0.55	8.5	70.3	62	1	0.81	-0.27
N27	CBP_NCBD x CBP_NCBD	negative	homo_negative	118	0.46	0.046	0.56	12.5	67.3	52	18	0.27	+2.70
N28	FOS_bZIP x FOS_bZIP	negative	homo_negative	124	0.31	0.006	0.51	20.7	74.4	61	53	0.66	-0.42
N29	P53_TAD x P53_TAD	negative	homo_negative	122	0.13	0.017	0.22	22.2	55.3	64	0	0.97	-2.29
N30	P21_full x P21_full	negative	homo_negative	328	0.24	0.042	0.27	27.6	46.6	148	2	0.81	+0.39
N31	EIF4EBP1 x EIF4EBP1	negative	homo_negative	236	0.14	0.019	0.24	27.1	48.2	111	0	0.66	+0.13
P01	PTMA_full x H10_full	positive	fuzzy	305	0.08	0.004	0.29	30.8	51.5	3	0	0.76	-0.72
P02	SRSF1_RS x U170K_RS	positive	fuzzy	259	0.08	0.015	0.33	30.0	73.6	0	0	1.00	-3.22
P03	G3BP1_IDR1 x G3BP1_RGG	positive	fuzzy	205	0.06	0.007	0.36	30.1	62.6	13	0	0.64	-2.98
P04	FMR1_CT x CAPRIN1_CT	positive	fuzzy	291	0.04	0.006	0.09	31.4	50.1	0	0	0.25	-1.80
P05	SNCA_full x TAU_2N4R	positive	fuzzy	581	0.07	0.005	0.12	31.7	53.4	2	0	0.85	+0.56
P06	NCOA3_ACTR x CBP_NCBD	positive	mutual_folding	106	0.70	0.021	0.72	6.3	76.9	66	48	0.37	-2.43
P07	FOS_bZIP x JUN_bZIP	positive	mutual_folding	129	0.74	0.005	0.71	8.5	86.6	66	55	0.60	-1.79
P08	FUS_LC x FUS_LC	positive	fuzzy	326	0.15	0.007	0.16	27.5	49.5	321	46	1.00	+4.81
P09	HNRNPA1_LC x HNRNPA1_LC	positive	fuzzy	270	0.21	0.026	0.21	26.1	50.8	392	81	1.00	+4.52
P10	HNRNPA2_LC x HNRNPA2_LC	positive	fuzzy	320	0.15	0.019	0.15	28.2	47.1	385	61	1.00	+4.52
P11	TDP43_CTD x TDP43_CTD	positive	fuzzy	296	0.12	0.010	0.17	28.4	44.9	305	28	1.00	+4.00
P12	DDX4_NIDR x DDX4_NIDR	positive	fuzzy	472	0.08	0.000	0.09	31.0	43.5	101	0	1.00	+1.45
P13	EWSR1_LC x EWSR1_LC	positive	fuzzy	530	0.13	0.006	0.14	30.1	47.5	301	30	1.00	+3.21

PAE = mean interchain PAE (Å). C sum = interchain contact-probability sum. C>.5 = interchain token pairs above 0.5. compos. = composition-baseline score (leave-one-out, pooled fit). sd = spread across the five samples.

Appendix A continued

ID	chains	label	class	res	ipTM	sd	pTM	PAE	pLDDT	C sum	C>.5	disor..	comp..
P14	TAF15_LC x TAF15_LC	positive	fuzzy	416	0.14	0.011	0.14	29.1	40.6	268	25	1.00	+4.13
P15	UBQLN2_CT x UBQLN2_CT	positive	fuzzy	350	0.17	0.010	0.20	28.9	58.3	203	23	0.59	+0.29
P16	NSP1_FG x NSP1_FG	positive	fuzzy	350	0.20	0.007	0.32	25.4	40.5	155	0	0.25	+3.91
P17	TAU_2N4R x TAU_2N4R	positive	fuzzy	882	0.23	0.034	0.23	30.3	44.1	291	84	0.96	-0.31
P18	SNCA_full x SNCA_full	positive	fuzzy	280	0.23	0.053	0.23	27.3	55.8	268	165	1.00	-0.49
P19	CAPRIN1_CT x CAPRIN1_CT	positive	fuzzy	206	0.09	0.017	0.11	28.2	49.2	92	0	1.00	+1.00
P20	MAX_bHLHZ x MAX_bHLHZ	positive	mutual_folding	162	0.80	0.005	0.80	6.1	86.8	90	89	0.31	-0.93
P21	GCN4_LZ x GCN4_LZ	positive	mutual_folding	66	0.73	0.027	0.76	4.1	88.1	45	45	1.00	-1.52
S01	PTMA_full x H10_full_scr	scramble	scramble	305	0.06	0.004	0.15	31.3	56.8	0	0	1.00	--
S03	G3BP1_IDR1 x G3BP1_RGG_scr	scramble	scramble	205	0.09	0.009	0.19	26.6	39.1	76	0	0.44	--
S04	FMR1_CT x CAPRIN1_CT_scr	scramble	scramble	291	0.12	0.016	0.15	29.4	43.2	83	0	0.42	--
S06	NCOA3_ACTR x CBP_NCBD_scr	scramble	scramble	106	0.46	0.068	0.44	17.5	64.1	52	2	0.78	--
S07	FOS_bZIP x JUN_bZIP_scr	scramble	scramble	129	0.40	0.008	0.57	16.7	77.9	60	29	0.62	--
S08	FUS_LC x FUS_LC_scr	scramble	scramble	326	0.05	0.005	0.08	31.4	39.5	0	0	0.75	--
S11	TDP43_CTD x TDP43_CTD_scr	scramble	scramble	296	0.10	0.008	0.15	29.3	44.6	51	0	0.45	--
S12	DDX4_NIDR x DDX4_NIDR_scr	scramble	scramble	472	0.06	0.000	0.13	31.6	41.7	0	0	1.00	--

PAE = mean interchain PAE (Å). C sum = interchain contact-probability sum. C>.5 = interchain token pairs above 0.5. compos. = composition-baseline score (leave-one-out, pooled fit). sd = spread across the five samples.

Appendix B - every group, every metric

31 groupings x 16 metrics = 496 cells, each with its own bootstrap interval, permutation p, BH-corrected q within the group, and delta against the composition baseline (464 cells have one). A PASS badge marks a delta interval clear of zero: 15 cells qualify, 14 of them in the memorised-control block and one outside it (fuzzy homotypic positives, contact probability sum). No pre-registered test passes.

Pre-registered / 1. positives vs real negatives (17 positive / 31 negative, length alone 0.676, composition 0.767)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	17/31	0.611	[0.45, 0.77]	0.2069	0.2758	-0.156	[-0.32, +0.01]
chain-pair ipTM	17/31	0.611	[0.45, 0.77]	0.2069	0.2758	-0.156	[-0.32, +0.01]
ranking score	17/31	0.567	[0.39, 0.74]	0.4488	0.5129	-0.199	[-0.39, -0.01]
pTM	17/31	0.472	[0.31, 0.64]	0.7603	0.7603	-0.294	[-0.51, -0.06]
fraction disordered	17/31	0.385	[0.23, 0.55]	0.1751	0.2758	-0.381	[-0.61, -0.13]
chain-pair PAE min	17/31	0.681	[0.51, 0.83]	0.0361	0.1000	-0.085	[-0.25, +0.08]
interchain PAE mean	17/31	0.556	[0.39, 0.72]	0.5238	0.5588	-0.211	[-0.39, -0.03]
interchain PAE p05	17/31	0.620	[0.46, 0.77]	0.1710	0.2758	-0.146	[-0.32, +0.02]
PAE frac < 10 A	17/31	0.617	[0.47, 0.77]	0.1322	0.2758	-0.150	[-0.30, +0.00]
PAE frac < 5 A	17/31	0.602	[0.47, 0.74]	0.1391	0.2758	-0.164	[-0.32, -0.00]
contact prob. sum	17/31	0.787	[0.62, 0.93]	0.0005	0.0080	+0.020	[-0.10, +0.14]
contact prob. density	17/31	0.605	[0.44, 0.76]	0.2283	0.2810	-0.161	[-0.31, -0.02]
contact prob. max	17/31	0.678	[0.51, 0.83]	0.0375	0.1000	-0.088	[-0.23, +0.05]
contacts > 0.5	17/31	0.672	[0.52, 0.82]	0.0181	0.0965	-0.095	[-0.26, +0.06]
contact particip. ratio	17/31	0.695	[0.54, 0.84]	0.0275	0.1000	-0.071	[-0.21, +0.07]
mean pLDDT	17/31	0.292	[0.14, 0.46]	0.0175	0.0965	-0.474	[-0.72, -0.21]

Pre-registered / 2a. mutual folding vs hard coiled-coil (4 positive / 3 negative, length alone 0.417, composition 0.250)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	4/3	0.417	[0.00, 0.88]	0.8601	1.0000	+0.167	[-0.50, +0.88]
chain-pair ipTM	4/3	0.417	[0.00, 0.88]	0.8601	1.0000	+0.167	[-0.50, +0.88]
ranking score	4/3	0.542	[0.00, 1.00]	0.9724	1.0000	+0.292	[-0.33, +0.88]
pTM	4/3	0.417	[0.00, 0.92]	0.8545	1.0000	+0.167	[-0.50, +0.75]
fraction disordered	4/3	0.500	[0.00, 1.00]	1.0000	1.0000	+0.250	[-0.42, +0.83]
chain-pair PAE min	4/3	0.625	[0.12, 1.00]	0.6856	1.0000	+0.375	[-0.42, +1.00]
interchain PAE mean	4/3	0.500	[0.00, 1.00]	1.0000	1.0000	+0.250	[-0.50, +1.00]
interchain PAE p05	4/3	0.583	[0.12, 1.00]	0.7446	1.0000	+0.333	[-0.50, +1.00]
PAE frac < 10 A	4/3	0.583	[0.00, 1.00]	0.7433	1.0000	+0.333	[-0.42, +1.00]
PAE frac < 5 A	4/3	0.667	[0.17, 1.00]	0.6316	1.0000	+0.417	[-0.25, +1.00]
contact prob. sum	4/3	0.417	[0.00, 0.83]	0.8557	1.0000	+0.167	[-0.50, +0.83]
contact prob. density	4/3	0.750	[0.25, 1.00]	0.3979	1.0000	+0.500	[-0.25, +1.00]
contact prob. max	4/3	0.500	[0.12, 0.88]	1.0000	1.0000	+0.250	[-0.38, +0.88]
contacts > 0.5	4/3	0.458	[0.00, 0.88]	0.9428	1.0000	+0.208	[-0.50, +0.88]
contact particip. ratio	4/3	0.667	[0.25, 1.00]	0.6306	1.0000	+0.417	[-0.25, +1.00]
mean pLDDT	4/3	0.167	[0.00, 0.50]	0.2235	1.0000	-0.083	[-0.75, +0.50]

Pre-registered / 2b. fuzzy positives vs real negatives (17 positive / 28 negative, length alone 0.641, composition 0.784)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	17/28	0.676	[0.51, 0.83]	0.0467	0.0721	-0.107	[-0.26, +0.05]
chain-pair ipTM	17/28	0.676	[0.51, 0.83]	0.0467	0.0721	-0.107	[-0.26, +0.05]
ranking score	17/28	0.628	[0.44, 0.80]	0.1538	0.1893	-0.155	[-0.35, +0.04]
pTM	17/28	0.523	[0.35, 0.70]	0.7950	0.7950	-0.261	[-0.49, -0.02]
fraction disordered	17/28	0.409	[0.25, 0.58]	0.2933	0.3128	-0.375	[-0.60, -0.13]
chain-pair PAE min	17/28	0.752	[0.59, 0.90]	0.0028	0.0148	-0.032	[-0.20, +0.12]
interchain PAE mean	17/28	0.616	[0.45, 0.78]	0.1991	0.2275	-0.168	[-0.34, +0.01]
interchain PAE p05	17/28	0.687	[0.52, 0.84]	0.0365	0.0721	-0.097	[-0.27, +0.07]
PAE frac < 10 A	17/28	0.683	[0.53, 0.83]	0.0147	0.0392	-0.101	[-0.25, +0.04]
PAE frac < 5 A	17/28	0.667	[0.53, 0.80]	0.0112	0.0358	-0.117	[-0.27, +0.04]
contact prob. sum	17/28	0.795	[0.62, 0.93]	0.0007	0.0112	+0.012	[-0.11, +0.13]
contact prob. density	17/28	0.655	[0.49, 0.81]	0.0785	0.1047	-0.128	[-0.27, +0.02]
contact prob. max	17/28	0.751	[0.58, 0.90]	0.0037	0.0148	-0.033	[-0.16, +0.09]
contacts > 0.5	17/28	0.721	[0.58, 0.86]	0.0021	0.0148	-0.063	[-0.21, +0.09]
contact particip. ratio	17/28	0.675	[0.51, 0.83]	0.0496	0.0721	-0.108	[-0.26, +0.04]
mean pLDDT	17/28	0.324	[0.16, 0.50]	0.0474	0.0721	-0.460	[-0.71, -0.17]

Appendix B continued

Pre-registered / 3. positives vs gold negatives (17 positive / 8 negative, length alone 0.985, composition 0.640)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	17/8	0.118	[0.00, 0.30]	0.0010	0.0035	-0.522	[-0.74, -0.29]
chain-pair ipTM	17/8	0.118	[0.00, 0.30]	0.0010	0.0035	-0.522	[-0.74, -0.29]
ranking score	17/8	0.239	[0.05, 0.48]	0.0374	0.0665	-0.401	[-0.68, -0.10]
pTM	17/8	0.074	[0.00, 0.21]	0.0004	0.0032	-0.566	[-0.79, -0.31]
fraction disordered	17/8	0.224	[0.06, 0.43]	0.0221	0.0442	-0.415	[-0.70, -0.10]
chain-pair PAE min	17/8	0.301	[0.11, 0.54]	0.1173	0.1444	-0.338	[-0.60, -0.08]
interchain PAE mean	17/8	0.015	[0.00, 0.07]	<0.0001	0.0016	-0.625	[-0.83, -0.41]
interchain PAE p05	17/8	0.110	[0.00, 0.26]	0.0013	0.0035	-0.529	[-0.73, -0.31]
PAE frac < 10 Å	17/8	0.265	[0.04, 0.53]	0.0593	0.0860	-0.375	[-0.65, -0.10]
PAE frac < 5 Å	17/8	0.276	[0.07, 0.53]	0.0576	0.0860	-0.364	[-0.64, -0.07]
contact prob. sum	17/8	0.691	[0.46, 0.88]	0.1387	0.1479	+0.051	[-0.15, +0.26]
contact prob. density	17/8	0.125	[0.01, 0.28]	0.0019	0.0043	-0.515	[-0.74, -0.29]
contact prob. max	17/8	0.265	[0.07, 0.51]	0.0645	0.0860	-0.375	[-0.64, -0.11]
contacts > 0.5	17/8	0.434	[0.21, 0.66]	0.6013	0.6013	-0.206	[-0.49, +0.07]
contact particip. ratio	17/8	0.691	[0.45, 0.91]	0.1371	0.1479	+0.051	[-0.25, +0.34]
mean pLDDT	17/8	0.118	[0.00, 0.29]	0.0012	0.0035	-0.522	[-0.79, -0.21]

Pre-registered / 4. positives vs scrambles (17 positive / 8 negative, length alone 0.684)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	17/8	0.566	[0.29, 0.83]	0.6202	0.7634	--	--
chain-pair ipTM	17/8	0.566	[0.29, 0.83]	0.6202	0.7634	--	--
ranking score	17/8	0.621	[0.33, 0.89]	0.3510	0.7397	--	--
pTM	17/8	0.511	[0.25, 0.77]	0.9441	0.9792	--	--
fraction disordered	17/8	0.309	[0.10, 0.54]	0.1088	0.3705	--	--
chain-pair PAE min	17/8	0.713	[0.49, 0.90]	0.0974	0.3705	--	--
interchain PAE mean	17/8	0.493	[0.21, 0.78]	0.9792	0.9792	--	--
interchain PAE p05	17/8	0.581	[0.30, 0.85]	0.5376	0.7634	--	--
PAE frac < 10 Å	17/8	0.596	[0.32, 0.85]	0.4374	0.7580	--	--
PAE frac < 5 Å	17/8	0.643	[0.42, 0.82]	0.2003	0.5341	--	--
contact prob. sum	17/8	0.783	[0.58, 0.94]	0.0215	0.3440	--	--
contact prob. density	17/8	0.544	[0.28, 0.80]	0.7644	0.8736	--	--
contact prob. max	17/8	0.728	[0.51, 0.90]	0.0718	0.3705	--	--
contacts > 0.5	17/8	0.684	[0.49, 0.85]	0.1158	0.3705	--	--
contact particip. ratio	17/8	0.614	[0.34, 0.87]	0.3699	0.7397	--	--
mean pLDDT	17/8	0.596	[0.29, 0.88]	0.4738	0.7580	--	--

By pair type / Homotypic only (12 positive / 11 negative, length alone 0.795, composition 0.902)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	12/11	0.564	[0.30, 0.83]	0.6081	0.7271	-0.337	[-0.58, -0.12]
chain-pair ipTM	12/11	0.564	[0.30, 0.83]	0.6081	0.7271	-0.337	[-0.58, -0.12]
ranking score	12/11	0.621	[0.38, 0.86]	0.3332	0.5331	-0.280	[-0.52, -0.05]
pTM	12/11	0.364	[0.13, 0.64]	0.2852	0.5070	-0.538	[-0.77, -0.29]
fraction disordered	12/11	0.295	[0.09, 0.52]	0.0771	0.2056	-0.606	[-0.83, -0.34]
chain-pair PAE min	12/11	0.795	[0.58, 0.95]	0.0162	0.0664	-0.106	[-0.32, +0.08]
interchain PAE mean	12/11	0.439	[0.19, 0.71]	0.6362	0.7271	-0.462	[-0.70, -0.21]
interchain PAE p05	12/11	0.591	[0.34, 0.83]	0.4623	0.6724	-0.311	[-0.56, -0.09]
PAE frac < 10 Å	12/11	0.682	[0.44, 0.91]	0.1381	0.3156	-0.220	[-0.46, +0.00]
PAE frac < 5 Å	12/11	0.644	[0.41, 0.87]	0.2184	0.4368	-0.258	[-0.49, -0.03]
contact prob. sum	12/11	0.970	[0.89, 1.00]	<0.0001	0.0016	+0.068	[-0.05, +0.22]
contact prob. density	12/11	0.530	[0.29, 0.78]	0.8364	0.8364	-0.371	[-0.63, -0.13]
contact prob. max	12/11	0.795	[0.58, 0.98]	0.0166	0.0664	-0.106	[-0.30, +0.05]
contacts > 0.5	12/11	0.792	[0.60, 0.95]	0.0130	0.0664	-0.110	[-0.31, +0.08]
contact particip. ratio	12/11	0.553	[0.30, 0.79]	0.6892	0.7352	-0.348	[-0.64, -0.05]
mean pLDDT	12/11	0.235	[0.06, 0.45]	0.0331	0.1059	-0.667	[-0.90, -0.39]

Appendix B continued

By pair type / Heterotypic only (5 positive / 20 negative, length alone 0.530, composition 0.500)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	5/20	0.485	[0.23, 0.74]	0.9390	1.0000	-0.015	[-0.32, +0.29]
chain-pair ipTM	5/20	0.485	[0.23, 0.74]	0.9390	1.0000	-0.015	[-0.32, +0.29]
ranking score	5/20	0.350	[0.10, 0.64]	0.3327	1.0000	-0.150	[-0.50, +0.22]
pTM	5/20	0.575	[0.25, 0.88]	0.6275	1.0000	+0.075	[-0.30, +0.43]
fraction disordered	5/20	0.550	[0.28, 0.81]	0.7357	1.0000	+0.050	[-0.26, +0.38]
chain-pair PAE min	5/20	0.420	[0.14, 0.72]	0.6003	1.0000	-0.080	[-0.42, +0.28]
interchain PAE mean	5/20	0.450	[0.19, 0.73]	0.7641	1.0000	-0.050	[-0.37, +0.26]
interchain PAE p05	5/20	0.460	[0.19, 0.75]	0.8026	1.0000	-0.040	[-0.38, +0.29]
PAE frac < 10 Å	5/20	0.375	[0.28, 0.47]	0.4519	1.0000	-0.125	[-0.34, +0.09]
PAE frac < 5 Å	5/20	0.425	[0.35, 0.50]	0.5870	1.0000	-0.075	[-0.29, +0.14]
contact prob. sum	5/20	0.465	[0.24, 0.70]	0.8287	1.0000	-0.035	[-0.29, +0.22]
contact prob. density	5/20	0.460	[0.23, 0.71]	0.8040	1.0000	-0.040	[-0.27, +0.21]
contact prob. max	5/20	0.485	[0.26, 0.72]	0.9303	1.0000	-0.015	[-0.27, +0.26]
contacts > 0.5	5/20	0.425	[0.35, 0.50]	0.5870	1.0000	-0.075	[-0.29, +0.14]
contact particip. ratio	5/20	0.585	[0.30, 0.85]	0.5729	1.0000	+0.085	[-0.20, +0.39]
mean pLDDT	5/20	0.500	[0.24, 0.77]	1.0000	1.0000	+0.000	[-0.32, +0.32]

By negative type / vs noncognate (17 positive / 10 negative, length alone 0.547, composition 0.753)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	17/10	0.768	[0.56, 0.94]	0.0179	0.0291	+0.015	[-0.24, +0.27]
chain-pair ipTM	17/10	0.768	[0.56, 0.94]	0.0179	0.0291	+0.015	[-0.24, +0.27]
ranking score	17/10	0.706	[0.49, 0.89]	0.0768	0.0945	-0.047	[-0.30, +0.21]
pTM	17/10	0.503	[0.26, 0.73]	0.9904	0.9904	-0.250	[-0.58, +0.08]
fraction disordered	17/10	0.341	[0.14, 0.55]	0.1619	0.1850	-0.412	[-0.69, -0.10]
chain-pair PAE min	17/10	0.806	[0.63, 0.95]	0.0070	0.0232	+0.053	[-0.16, +0.27]
interchain PAE mean	17/10	0.759	[0.56, 0.94]	0.0279	0.0406	+0.006	[-0.19, +0.22]
interchain PAE p05	17/10	0.788	[0.60, 0.95]	0.0116	0.0265	+0.035	[-0.19, +0.25]
PAE frac < 10 Å	17/10	0.724	[0.55, 0.88]	0.0372	0.0496	-0.029	[-0.24, +0.19]
PAE frac < 5 Å	17/10	0.735	[0.62, 0.85]	0.0182	0.0291	-0.018	[-0.21, +0.19]
contact prob. sum	17/10	0.829	[0.66, 0.96]	0.0037	0.0232	+0.076	[-0.10, +0.26]
contact prob. density	17/10	0.812	[0.64, 0.95]	0.0064	0.0232	+0.059	[-0.10, +0.23]
contact prob. max	17/10	0.838	[0.67, 0.97]	0.0031	0.0232	+0.085	[-0.09, +0.28]
contacts > 0.5	17/10	0.765	[0.65, 0.88]	0.0087	0.0232	+0.012	[-0.17, +0.21]
contact particip. ratio	17/10	0.800	[0.61, 0.95]	0.0085	0.0232	+0.047	[-0.14, +0.25]
mean pLDDT	17/10	0.365	[0.16, 0.59]	0.2538	0.2707	-0.388	[-0.66, -0.08]

By negative type / vs cross_kingdom (17 positive / 5 negative, length alone 0.412, composition 0.894)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	17/5	0.847	[0.66, 0.99]	0.0172	0.0393	-0.047	[-0.19, +0.09]
chain-pair ipTM	17/5	0.847	[0.66, 0.99]	0.0172	0.0393	-0.047	[-0.19, +0.09]
ranking score	17/5	0.618	[0.39, 0.82]	0.4554	0.4554	-0.276	[-0.49, -0.06]
pTM	17/5	0.694	[0.42, 0.94]	0.2091	0.2230	-0.200	[-0.48, +0.05]
fraction disordered	17/5	0.706	[0.59, 0.82]	0.1649	0.1884	-0.188	[-0.36, +0.02]
chain-pair PAE min	17/5	0.871	[0.71, 1.00]	0.0111	0.0358	-0.024	[-0.19, +0.13]
interchain PAE mean	17/5	0.824	[0.55, 1.00]	0.0295	0.0524	-0.071	[-0.31, +0.14]
interchain PAE p05	17/5	0.871	[0.69, 1.00]	0.0112	0.0358	-0.024	[-0.19, +0.14]
PAE frac < 10 Å	17/5	0.794	[0.68, 0.91]	0.0498	0.0797	-0.100	[-0.24, +0.04]
PAE frac < 5 Å	17/5	0.735	[0.62, 0.85]	0.1170	0.1440	-0.159	[-0.30, -0.01]
contact prob. sum	17/5	0.900	[0.74, 1.00]	0.0049	0.0358	+0.006	[-0.09, +0.12]
contact prob. density	17/5	0.824	[0.56, 1.00]	0.0262	0.0524	-0.071	[-0.28, +0.10]
contact prob. max	17/5	0.906	[0.75, 1.00]	0.0039	0.0358	+0.012	[-0.09, +0.13]
contacts > 0.5	17/5	0.765	[0.65, 0.88]	0.0825	0.1100	-0.129	[-0.26, +0.02]
contact particip. ratio	17/5	0.876	[0.68, 1.00]	0.0094	0.0358	-0.018	[-0.15, +0.12]
mean pLDDT	17/5	0.224	[0.05, 0.42]	0.0743	0.1081	-0.671	[-0.88, -0.40]

Appendix B continued

By negative type / vs charge_repulsive (17 positive / 2 negative, length alone 0.941, composition 0.853)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	17/2	0.956	[0.82, 1.00]	n/a	n/a	+0.103	[-0.03, +0.29]
chain-pair ipTM	17/2	0.956	[0.82, 1.00]	n/a	n/a	+0.103	[-0.03, +0.29]
ranking score	17/2	0.941	[0.82, 1.00]	n/a	n/a	+0.088	[-0.09, +0.29]
pTM	17/2	0.882	[0.71, 1.00]	n/a	n/a	+0.029	[-0.18, +0.24]
fraction disordered	17/2	0.118	[0.00, 0.29]	n/a	n/a	-0.735	[-0.94, -0.47]
chain-pair PAE min	17/2	0.882	[0.71, 1.00]	n/a	n/a	+0.029	[-0.18, +0.24]
interchain PAE mean	17/2	0.794	[0.59, 0.97]	n/a	n/a	-0.059	[-0.35, +0.21]
interchain PAE p05	17/2	0.882	[0.71, 1.00]	n/a	n/a	+0.029	[-0.18, +0.24]
PAE frac < 10 Å	17/2	0.794	[0.68, 0.91]	n/a	n/a	-0.059	[-0.21, +0.09]
PAE frac < 5 Å	17/2	0.735	[0.62, 0.85]	n/a	n/a	-0.118	[-0.26, +0.06]
contact prob. sum	17/2	0.971	[0.91, 1.00]	n/a	n/a	+0.118	[-0.03, +0.29]
contact prob. density	17/2	0.941	[0.85, 1.00]	n/a	n/a	+0.088	[-0.03, +0.24]
contact prob. max	17/2	0.971	[0.91, 1.00]	n/a	n/a	+0.118	[-0.03, +0.29]
contacts > 0.5	17/2	0.765	[0.65, 0.88]	n/a	n/a	-0.088	[-0.24, +0.06]
contact particip. ratio	17/2	0.971	[0.91, 1.00]	n/a	n/a	+0.118	[-0.03, +0.29]
mean pLDDT	17/2	0.294	[0.06, 0.59]	n/a	n/a	-0.559	[-0.88, -0.18]

By negative type / vs hard_coiledcoil (17 positive / 3 negative, length alone 1.000, composition 0.608)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	17/3	0.000	[0.00, 0.00]	0.0021	0.0034	-0.608	[-0.88, -0.29]
chain-pair ipTM	17/3	0.000	[0.00, 0.00]	0.0021	0.0034	-0.608	[-0.88, -0.29]
ranking score	17/3	0.000	[0.00, 0.00]	0.0017	0.0034	-0.608	[-0.88, -0.29]
pTM	17/3	0.000	[0.00, 0.00]	0.0020	0.0034	-0.608	[-0.88, -0.29]
fraction disordered	17/3	0.167	[0.00, 0.35]	0.0352	0.0469	-0.441	[-0.82, +0.00]
chain-pair PAE min	17/3	0.020	[0.00, 0.12]	0.0024	0.0035	-0.588	[-0.88, -0.25]
interchain PAE mean	17/3	0.000	[0.00, 0.00]	0.0017	0.0034	-0.608	[-0.88, -0.29]
interchain PAE p05	17/3	0.000	[0.00, 0.00]	0.0017	0.0034	-0.608	[-0.88, -0.29]
PAE frac < 10 Å	17/3	0.000	[0.00, 0.00]	0.0010	0.0034	-0.608	[-0.88, -0.29]
PAE frac < 5 Å	17/3	0.000	[0.00, 0.00]	0.0010	0.0034	-0.608	[-0.88, -0.29]
contact prob. sum	17/3	0.706	[0.47, 0.88]	0.2817	0.2817	+0.098	[-0.18, +0.37]
contact prob. density	17/3	0.137	[0.00, 0.33]	0.0574	0.0656	-0.471	[-0.76, -0.16]
contact prob. max	17/3	0.000	[0.00, 0.00]	0.0017	0.0034	-0.608	[-0.88, -0.29]
contacts > 0.5	17/3	0.216	[0.06, 0.41]	0.1157	0.1234	-0.392	[-0.76, +0.00]
contact particip. ratio	17/3	0.882	[0.71, 1.00]	0.0390	0.0480	+0.275	[+0.00, +0.59]
mean pLDDT	17/3	0.000	[0.00, 0.00]	0.0019	0.0034	-0.608	[-0.88, -0.29]

By negative type / vs homo_negative (17 positive / 11 negative, length alone 0.775, composition 0.749)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	17/11	0.465	[0.22, 0.71]	0.7738	0.8844	-0.283	[-0.49, -0.10]
chain-pair ipTM	17/11	0.465	[0.22, 0.71]	0.7738	0.8844	-0.283	[-0.49, -0.10]
ranking score	17/11	0.505	[0.29, 0.73]	0.9730	0.9730	-0.243	[-0.48, +0.00]
pTM	17/11	0.398	[0.18, 0.64]	0.3726	0.7451	-0.350	[-0.61, -0.08]
fraction disordered	17/11	0.388	[0.19, 0.59]	0.3048	0.7451	-0.361	[-0.62, -0.08]
chain-pair PAE min	17/11	0.626	[0.41, 0.82]	0.2876	0.7451	-0.123	[-0.32, +0.06]
interchain PAE mean	17/11	0.358	[0.14, 0.60]	0.2269	0.7451	-0.390	[-0.61, -0.16]
interchain PAE p05	17/11	0.476	[0.25, 0.71]	0.8526	0.9095	-0.273	[-0.49, -0.06]
PAE frac < 10 Å	17/11	0.575	[0.35, 0.78]	0.4940	0.7903	-0.174	[-0.37, +0.01]
PAE frac < 5 Å	17/11	0.561	[0.35, 0.76]	0.5622	0.8019	-0.187	[-0.39, +0.01]
contact prob. sum	17/11	0.684	[0.47, 0.88]	0.1114	0.7451	-0.064	[-0.24, +0.11]
contact prob. density	17/11	0.385	[0.18, 0.61]	0.3268	0.7451	-0.364	[-0.57, -0.16]
contact prob. max	17/11	0.561	[0.34, 0.79]	0.6014	0.8019	-0.187	[-0.37, +0.01]
contacts > 0.5	17/11	0.652	[0.47, 0.82]	0.1514	0.7451	-0.096	[-0.29, +0.10]
contact particip. ratio	17/11	0.417	[0.21, 0.64]	0.4893	0.7903	-0.332	[-0.56, -0.09]
mean pLDDT	17/11	0.337	[0.14, 0.56]	0.1636	0.7451	-0.412	[-0.71, -0.06]

Appendix B continued

By negative type / vs scramble (17 positive / 8 negative, length alone 0.684)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	17/8	0.566	[0.29, 0.83]	0.6202	0.7634	--	--
chain-pair ipTM	17/8	0.566	[0.29, 0.83]	0.6202	0.7634	--	--
ranking score	17/8	0.621	[0.33, 0.89]	0.3510	0.7397	--	--
pTM	17/8	0.511	[0.25, 0.77]	0.9441	0.9792	--	--
fraction disordered	17/8	0.309	[0.10, 0.54]	0.1088	0.3705	--	--
chain-pair PAE min	17/8	0.713	[0.49, 0.90]	0.0974	0.3705	--	--
interchain PAE mean	17/8	0.493	[0.21, 0.78]	0.9792	0.9792	--	--
interchain PAE p05	17/8	0.581	[0.30, 0.85]	0.5376	0.7634	--	--
PAE frac < 10 Å	17/8	0.596	[0.32, 0.85]	0.4374	0.7580	--	--
PAE frac < 5 Å	17/8	0.643	[0.42, 0.82]	0.2003	0.5341	--	--
contact prob. sum	17/8	0.783	[0.58, 0.94]	0.0215	0.3440	--	--
contact prob. density	17/8	0.544	[0.28, 0.80]	0.7644	0.8736	--	--
contact prob. max	17/8	0.728	[0.51, 0.90]	0.0718	0.3705	--	--
contacts > 0.5	17/8	0.684	[0.49, 0.85]	0.1158	0.3705	--	--
contact particip. ratio	17/8	0.614	[0.34, 0.87]	0.3699	0.7397	--	--
mean pLDDT	17/8	0.596	[0.29, 0.88]	0.4738	0.7580	--	--

By negative evidence / vs demonstrated (17 positive / 5 negative, length alone 0.976, composition 0.671)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	17/5	0.188	[0.00, 0.45]	0.0337	0.0770	-0.482	[-0.74, -0.21]
chain-pair ipTM	17/5	0.188	[0.00, 0.45]	0.0337	0.0770	-0.482	[-0.74, -0.21]
ranking score	17/5	0.241	[0.00, 0.56]	0.0836	0.1486	-0.429	[-0.74, -0.08]
pTM	17/5	0.118	[0.00, 0.31]	0.0076	0.0581	-0.553	[-0.81, -0.26]
fraction disordered	17/5	0.229	[0.06, 0.45]	0.0576	0.1152	-0.441	[-0.74, -0.11]
chain-pair PAE min	17/5	0.247	[0.00, 0.53]	0.0984	0.1574	-0.424	[-0.71, -0.13]
interchain PAE mean	17/5	0.024	[0.00, 0.11]	<0.0001	0.0016	-0.647	[-0.88, -0.40]
interchain PAE p05	17/5	0.141	[0.00, 0.35]	0.0109	0.0581	-0.529	[-0.76, -0.28]
PAE frac < 10 Å	17/5	0.318	[0.00, 0.68]	0.2320	0.2855	-0.353	[-0.69, -0.01]
PAE frac < 5 Å	17/5	0.294	[0.00, 0.64]	0.1598	0.2130	-0.376	[-0.71, -0.05]
contact prob. sum	17/5	0.682	[0.46, 0.88]	0.2520	0.2880	+0.012	[-0.22, +0.26]
contact prob. density	17/5	0.165	[0.01, 0.36]	0.0230	0.0770	-0.506	[-0.78, -0.22]
contact prob. max	17/5	0.271	[0.00, 0.61]	0.1279	0.1860	-0.400	[-0.71, -0.09]
contacts > 0.5	17/5	0.435	[0.15, 0.71]	0.6840	0.6840	-0.235	[-0.56, +0.08]
contact particip. ratio	17/5	0.635	[0.31, 0.94]	0.4024	0.4292	-0.035	[-0.41, +0.35]
mean pLDDT	17/5	0.165	[0.00, 0.42]	0.0271	0.0770	-0.506	[-0.81, -0.14]

By negative evidence / vs demonstrated+matched_internal (17 positive / 3 negative, length alone 1.000, composition 0.588)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	17/3	0.000	[0.00, 0.00]	0.0021	0.0084	-0.588	[-0.82, -0.33]
chain-pair ipTM	17/3	0.000	[0.00, 0.00]	0.0021	0.0084	-0.588	[-0.82, -0.33]
ranking score	17/3	0.235	[0.00, 0.55]	0.1559	0.2395	-0.353	[-0.76, +0.12]
pTM	17/3	0.000	[0.00, 0.00]	0.0020	0.0084	-0.588	[-0.82, -0.33]
fraction disordered	17/3	0.216	[0.04, 0.43]	0.1647	0.2395	-0.373	[-0.69, -0.04]
chain-pair PAE min	17/3	0.392	[0.14, 0.65]	0.5820	0.6208	-0.196	[-0.49, +0.08]
interchain PAE mean	17/3	0.000	[0.00, 0.00]	0.0017	0.0084	-0.588	[-0.82, -0.33]
interchain PAE p05	17/3	0.059	[0.00, 0.24]	0.0138	0.0315	-0.529	[-0.76, -0.27]
PAE frac < 10 Å	17/3	0.176	[0.00, 0.51]	0.0772	0.1544	-0.412	[-0.71, -0.12]
PAE frac < 5 Å	17/3	0.245	[0.00, 0.71]	0.1830	0.2440	-0.343	[-0.71, +0.04]
contact prob. sum	17/3	0.706	[0.47, 0.88]	0.2814	0.3216	+0.118	[-0.12, +0.35]
contact prob. density	17/3	0.059	[0.00, 0.20]	0.0111	0.0296	-0.529	[-0.80, -0.24]
contact prob. max	17/3	0.255	[0.00, 0.59]	0.2165	0.2664	-0.333	[-0.71, +0.06]
contacts > 0.5	17/3	0.431	[0.18, 0.71]	0.7483	0.7483	-0.157	[-0.53, +0.20]
contact particip. ratio	17/3	0.784	[0.57, 0.96]	0.1425	0.2395	+0.196	[-0.04, +0.47]
mean pLDDT	17/3	0.039	[0.00, 0.16]	0.0063	0.0202	-0.549	[-0.82, -0.24]

Appendix B continued

By negative evidence / vs gold (both demonstrated grades) (17 positive / 8 negative, length alone 0.985, composition 0.640)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	17/8	0.118	[0.00, 0.30]	0.0010	0.0035	-0.522	[-0.74, -0.29]
chain-pair ipTM	17/8	0.118	[0.00, 0.30]	0.0010	0.0035	-0.522	[-0.74, -0.29]
ranking score	17/8	0.239	[0.05, 0.48]	0.0374	0.0665	-0.401	[-0.68, -0.10]
pTM	17/8	0.074	[0.00, 0.21]	0.0004	0.0032	-0.566	[-0.79, -0.31]
fraction disordered	17/8	0.224	[0.06, 0.43]	0.0221	0.0442	-0.415	[-0.70, -0.10]
chain-pair PAE min	17/8	0.301	[0.11, 0.54]	0.1173	0.1444	-0.338	[-0.60, -0.08]
interchain PAE mean	17/8	0.015	[0.00, 0.07]	<0.0001	0.0016	-0.625	[-0.83, -0.41]
interchain PAE p05	17/8	0.110	[0.00, 0.26]	0.0013	0.0035	-0.529	[-0.73, -0.31]
PAE frac < 10 A	17/8	0.265	[0.04, 0.53]	0.0593	0.0860	-0.375	[-0.65, -0.10]
PAE frac < 5 A	17/8	0.276	[0.07, 0.53]	0.0576	0.0860	-0.364	[-0.64, -0.07]
contact prob. sum	17/8	0.691	[0.46, 0.88]	0.1387	0.1479	+0.051	[-0.15, +0.26]
contact prob. density	17/8	0.125	[0.01, 0.28]	0.0019	0.0043	-0.515	[-0.74, -0.29]
contact prob. max	17/8	0.265	[0.07, 0.51]	0.0645	0.0860	-0.375	[-0.64, -0.11]
contacts > 0.5	17/8	0.434	[0.21, 0.66]	0.6013	0.6013	-0.206	[-0.49, +0.07]
contact particip. ratio	17/8	0.691	[0.45, 0.91]	0.1371	0.1479	+0.051	[-0.25, +0.34]
mean pLDDT	17/8	0.118	[0.00, 0.29]	0.0012	0.0035	-0.522	[-0.79, -0.21]

By negative evidence / vs inferred (17 positive / 11 negative, length alone 0.647, composition 0.872)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	17/11	0.805	[0.59, 0.98]	0.0057	0.0182	-0.067	[-0.22, +0.07]
chain-pair ipTM	17/11	0.805	[0.59, 0.98]	0.0057	0.0182	-0.067	[-0.22, +0.07]
ranking score	17/11	0.652	[0.44, 0.85]	0.1833	0.1955	-0.219	[-0.44, -0.02]
pTM	17/11	0.727	[0.51, 0.92]	0.0448	0.0672	-0.144	[-0.37, +0.07]
fraction disordered	17/11	0.532	[0.34, 0.71]	0.7686	0.7686	-0.340	[-0.57, -0.10]
chain-pair PAE min	17/11	0.813	[0.63, 0.97]	0.0045	0.0182	-0.059	[-0.23, +0.10]
interchain PAE mean	17/11	0.711	[0.50, 0.89]	0.0628	0.0837	-0.160	[-0.35, +0.02]
interchain PAE p05	17/11	0.791	[0.61, 0.95]	0.0086	0.0229	-0.080	[-0.25, +0.08]
PAE frac < 10 A	17/11	0.743	[0.59, 0.88]	0.0174	0.0348	-0.128	[-0.26, -0.00]
PAE frac < 5 A	17/11	0.695	[0.54, 0.82]	0.0462	0.0672	-0.176	[-0.32, -0.04]
contact prob. sum	17/11	0.832	[0.66, 0.96]	0.0027	0.0182	-0.040	[-0.17, +0.08]
contact prob. density	17/11	0.738	[0.53, 0.90]	0.0326	0.0579	-0.134	[-0.31, +0.02]
contact prob. max	17/11	0.829	[0.66, 0.95]	0.0028	0.0182	-0.043	[-0.18, +0.08]
contacts > 0.5	17/11	0.743	[0.60, 0.87]	0.0118	0.0270	-0.128	[-0.27, +0.01]
contact particip. ratio	17/11	0.703	[0.48, 0.90]	0.0705	0.0868	-0.168	[-0.36, +0.01]
mean pLDDT	17/11	0.305	[0.11, 0.51]	0.0938	0.1072	-0.567	[-0.83, -0.27]

By negative evidence / vs absence_only (17 positive / 10 negative, length alone 0.547, composition 0.753)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	17/10	0.768	[0.56, 0.94]	0.0179	0.0291	+0.015	[-0.24, +0.27]
chain-pair ipTM	17/10	0.768	[0.56, 0.94]	0.0179	0.0291	+0.015	[-0.24, +0.27]
ranking score	17/10	0.706	[0.49, 0.89]	0.0768	0.0945	-0.047	[-0.30, +0.21]
pTM	17/10	0.503	[0.26, 0.73]	0.9904	0.9904	-0.250	[-0.58, +0.08]
fraction disordered	17/10	0.341	[0.14, 0.55]	0.1619	0.1850	-0.412	[-0.69, -0.10]
chain-pair PAE min	17/10	0.806	[0.63, 0.95]	0.0070	0.0232	+0.053	[-0.16, +0.27]
interchain PAE mean	17/10	0.759	[0.56, 0.94]	0.0279	0.0406	+0.006	[-0.19, +0.22]
interchain PAE p05	17/10	0.788	[0.60, 0.95]	0.0116	0.0265	+0.035	[-0.19, +0.25]
PAE frac < 10 A	17/10	0.724	[0.55, 0.88]	0.0372	0.0496	-0.029	[-0.24, +0.19]
PAE frac < 5 A	17/10	0.735	[0.62, 0.85]	0.0182	0.0291	-0.018	[-0.21, +0.19]
contact prob. sum	17/10	0.829	[0.66, 0.96]	0.0037	0.0232	+0.076	[-0.10, +0.26]
contact prob. density	17/10	0.812	[0.64, 0.95]	0.0064	0.0232	+0.059	[-0.10, +0.23]
contact prob. max	17/10	0.838	[0.67, 0.97]	0.0031	0.0232	+0.085	[-0.09, +0.28]
contacts > 0.5	17/10	0.765	[0.65, 0.88]	0.0087	0.0232	+0.012	[-0.17, +0.21]
contact particip. ratio	17/10	0.800	[0.61, 0.95]	0.0085	0.0232	+0.047	[-0.14, +0.25]
mean pLDDT	17/10	0.365	[0.16, 0.59]	0.2538	0.2707	-0.388	[-0.66, -0.08]

Appendix B continued

By negative evidence / vs uncertain (17 positive / 2 negative, length alone 0.235, composition 0.765)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	17/2	0.735	[0.47, 0.94]	n/a	n/a	-0.029	[-0.24, +0.15]
chain-pair ipTM	17/2	0.735	[0.47, 0.94]	n/a	n/a	-0.029	[-0.24, +0.15]
ranking score	17/2	0.721	[0.41, 1.00]	n/a	n/a	-0.044	[-0.32, +0.21]
pTM	17/2	0.515	[0.12, 0.91]	n/a	n/a	-0.250	[-0.76, +0.29]
fraction disordered	17/2	0.441	[0.06, 0.79]	n/a	n/a	-0.324	[-0.82, +0.18]
chain-pair PAE min	17/2	0.853	[0.65, 1.00]	n/a	n/a	+0.088	[-0.12, +0.29]
interchain PAE mean	17/2	0.853	[0.65, 1.00]	n/a	n/a	+0.088	[-0.18, +0.35]
interchain PAE p05	17/2	0.882	[0.71, 1.00]	n/a	n/a	+0.118	[-0.12, +0.35]
PAE frac < 10 A	17/2	0.794	[0.68, 0.91]	n/a	n/a	+0.029	[-0.15, +0.24]
PAE frac < 5 A	17/2	0.735	[0.62, 0.85]	n/a	n/a	-0.029	[-0.21, +0.18]
contact prob. sum	17/2	0.706	[0.47, 0.88]	n/a	n/a	-0.059	[-0.24, +0.12]
contact prob. density	17/2	0.765	[0.53, 0.94]	n/a	n/a	+0.000	[-0.24, +0.24]
contact prob. max	17/2	0.706	[0.47, 0.88]	n/a	n/a	-0.059	[-0.24, +0.12]
contacts > 0.5	17/2	0.765	[0.65, 0.88]	n/a	n/a	+0.000	[-0.18, +0.21]
contact particip. ratio	17/2	0.147	[0.00, 0.35]	n/a	n/a	-0.618	[-0.82, -0.38]
mean pLDDT	17/2	0.559	[0.18, 0.94]	n/a	n/a	-0.206	[-0.76, +0.29]

By negative evidence / vs weak (inferred + absence_only) (17 positive / 21 negative, length alone 0.599, composition 0.815)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	17/21	0.787	[0.62, 0.93]	0.0012	0.0032	-0.028	[-0.18, +0.14]
chain-pair ipTM	17/21	0.787	[0.62, 0.93]	0.0012	0.0032	-0.028	[-0.18, +0.14]
ranking score	17/21	0.678	[0.49, 0.85]	0.0625	0.0769	-0.137	[-0.33, +0.06]
pTM	17/21	0.620	[0.43, 0.80]	0.2165	0.2309	-0.195	[-0.43, +0.04]
fraction disordered	17/21	0.441	[0.27, 0.61]	0.5124	0.5124	-0.374	[-0.59, -0.14]
chain-pair PAE min	17/21	0.810	[0.65, 0.95]	0.0009	0.0032	-0.006	[-0.17, +0.15]
interchain PAE mean	17/21	0.734	[0.56, 0.88]	0.0120	0.0160	-0.081	[-0.25, +0.09]
interchain PAE p05	17/21	0.790	[0.63, 0.93]	0.0022	0.0044	-0.025	[-0.20, +0.14]
PAE frac < 10 A	17/21	0.734	[0.59, 0.87]	0.0020	0.0044	-0.081	[-0.23, +0.06]
PAE frac < 5 A	17/21	0.714	[0.59, 0.84]	0.0026	0.0045	-0.101	[-0.25, +0.06]
contact prob. sum	17/21	0.831	[0.68, 0.95]	0.0002	0.0016	+0.015	[-0.11, +0.14]
contact prob. density	17/21	0.773	[0.61, 0.91]	0.0028	0.0045	-0.042	[-0.18, +0.09]
contact prob. max	17/21	0.833	[0.69, 0.95]	0.0003	0.0016	+0.018	[-0.11, +0.15]
contacts > 0.5	17/21	0.754	[0.63, 0.88]	0.0002	0.0016	-0.062	[-0.21, +0.09]
contact particip. ratio	17/21	0.749	[0.58, 0.90]	0.0063	0.0092	-0.066	[-0.21, +0.08]
mean pLDDT	17/21	0.333	[0.16, 0.52]	0.0791	0.0904	-0.482	[-0.74, -0.19]

By positive subgroup / Fuzzy heterotypic positives (5 positive / 31 negative, length alone 0.600, composition 0.452)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	5/31	0.394	[0.19, 0.61]	0.4642	0.6958	-0.058	[-0.32, +0.21]
chain-pair ipTM	5/31	0.394	[0.19, 0.61]	0.4642	0.6958	-0.058	[-0.32, +0.21]
ranking score	5/31	0.306	[0.09, 0.57]	0.1806	0.6958	-0.145	[-0.50, +0.24]
pTM	5/31	0.542	[0.24, 0.83]	0.7744	0.8261	+0.090	[-0.35, +0.51]
fraction disordered	5/31	0.571	[0.31, 0.82]	0.6312	0.7214	+0.119	[-0.23, +0.46]
chain-pair PAE min	5/31	0.348	[0.12, 0.61]	0.3053	0.6958	-0.103	[-0.54, +0.26]
interchain PAE mean	5/31	0.348	[0.14, 0.57]	0.3087	0.6958	-0.103	[-0.52, +0.26]
interchain PAE p05	5/31	0.368	[0.14, 0.61]	0.3666	0.6958	-0.084	[-0.52, +0.30]
PAE frac < 10 A	5/31	0.355	[0.27, 0.44]	0.3058	0.6958	-0.097	[-0.34, +0.13]
PAE frac < 5 A	5/31	0.403	[0.34, 0.47]	0.5653	0.6958	-0.048	[-0.29, +0.18]
contact prob. sum	5/31	0.300	[0.14, 0.48]	0.1586	0.6958	-0.152	[-0.41, +0.11]
contact prob. density	5/31	0.310	[0.15, 0.50]	0.1775	0.6958	-0.142	[-0.39, +0.12]
contact prob. max	5/31	0.313	[0.15, 0.50]	0.1903	0.6958	-0.139	[-0.41, +0.14]
contacts > 0.5	5/31	0.387	[0.31, 0.45]	0.5003	0.6958	-0.065	[-0.30, +0.16]
contact particip. ratio	5/31	0.410	[0.20, 0.64]	0.5360	0.6958	-0.042	[-0.32, +0.25]
mean pLDDT	5/31	0.529	[0.29, 0.77]	0.8589	0.8589	+0.077	[-0.30, +0.46]

Appendix B continued

By positive subgroup / Fuzzy homotypic positives (12 positive / 31 negative, length alone 0.707, composition 0.898)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI	
ipTM	12/31	0.702	[0.54, 0.85]	0.0406	0.0509	-0.196	[-0.37, -0.03]	
chain-pair ipTM	12/31	0.702	[0.54, 0.85]	0.0406	0.0509	-0.196	[-0.37, -0.03]	
ranking score	12/31	0.676	[0.49, 0.84]	0.0782	0.0894	-0.222	[-0.43, -0.02]	
pTM	12/31	0.444	[0.27, 0.62]	0.5807	0.5807	-0.454	[-0.64, -0.26]	
fraction disordered	12/31	0.308	[0.15, 0.49]	0.0414	0.0509	-0.590	[-0.77, -0.39]	
chain-pair PAE min	12/31	0.820	[0.69, 0.93]	0.0017	0.0045	-0.078	[-0.23, +0.08]	
interchain PAE mean	12/31	0.642	[0.48, 0.80]	0.1570	0.1674	-0.255	[-0.42, -0.09]	
interchain PAE p05	12/31	0.726	[0.57, 0.87]	0.0222	0.0386	-0.172	[-0.33, -0.01]	
PAE frac < 10 A	12/31	0.726	[0.57, 0.87]	0.0108	0.0247	-0.172	[-0.35, +0.01]	
PAE frac < 5 A	12/31	0.685	[0.53, 0.84]	0.0241	0.0386	-0.212	[-0.40, -0.02]	
contact prob. sum	12/31	0.989	[0.96, 1.00]	<0.0001	0.0016	+0.091	[+0.01, +0.20]	● PASS
contact prob. density	12/31	0.728	[0.58, 0.87]	0.0200	0.0386	-0.169	[-0.33, -0.02]	
contact prob. max	12/31	0.831	[0.70, 0.95]	0.0009	0.0036	-0.067	[-0.22, +0.09]	
contacts > 0.5	12/31	0.790	[0.63, 0.93]	0.0009	0.0036	-0.108	[-0.30, +0.09]	
contact particip. ratio	12/31	0.815	[0.68, 0.93]	0.0008	0.0036	-0.083	[-0.22, +0.05]	
mean pLDDT	12/31	0.194	[0.06, 0.36]	0.0016	0.0045	-0.704	[-0.89, -0.47]	

By positive subgroup / Mutual-folding (memorised) positives (4 positive / 31 negative, length alone 0.129, composition 0.419)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI	
ipTM	4/31	0.944	[0.86, 1.00]	0.0012	0.0017	+0.524	[+0.34, +0.70]	● PASS
chain-pair ipTM	4/31	0.944	[0.86, 1.00]	0.0012	0.0017	+0.524	[+0.34, +0.70]	● PASS
ranking score	4/31	0.948	[0.85, 1.00]	0.0010	0.0017	+0.528	[+0.34, +0.70]	● PASS
pTM	4/31	0.944	[0.85, 1.00]	0.0010	0.0017	+0.524	[+0.35, +0.69]	● PASS
fraction disordered	4/31	0.665	[0.34, 0.92]	0.2974	0.3172	+0.246	[-0.13, +0.57]	
chain-pair PAE min	4/31	0.964	[0.89, 1.00]	0.0003	0.0012	+0.544	[+0.35, +0.73]	● PASS
interchain PAE mean	4/31	0.944	[0.85, 1.00]	0.0010	0.0017	+0.524	[+0.33, +0.71]	● PASS
interchain PAE p05	4/31	0.960	[0.88, 1.00]	<0.0001	0.0012	+0.540	[+0.34, +0.73]	● PASS
PAE frac < 10 A	4/31	0.952	[0.86, 1.00]	0.0005	0.0016	+0.532	[+0.34, +0.72]	● PASS
PAE frac < 5 A	4/31	0.968	[0.90, 1.00]	0.0002	0.0012	+0.548	[+0.36, +0.73]	● PASS
contact prob. sum	4/31	0.847	[0.69, 0.97]	0.0212	0.0242	+0.427	[+0.23, +0.63]	● PASS
contact prob. density	4/31	0.919	[0.79, 1.00]	0.0043	0.0053	+0.500	[+0.27, +0.71]	● PASS
contact prob. max	4/31	0.952	[0.87, 1.00]	0.0003	0.0012	+0.532	[+0.35, +0.71]	● PASS
contacts > 0.5	4/31	0.931	[0.83, 1.00]	0.0011	0.0017	+0.512	[+0.33, +0.69]	● PASS
contact particip. ratio	4/31	0.419	[0.25, 0.60]	0.6346	0.6346	+0.000	[-0.22, +0.23]	
mean pLDDT	4/31	0.919	[0.81, 1.00]	0.0026	0.0035	+0.500	[+0.30, +0.69]	● PASS

By positive subgroup / All 21 positives incl. memorised (21 positive / 31 negative, length alone 0.571, composition 0.700)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI	
ipTM	21/31	0.674	[0.52, 0.81]	0.0361	0.0642	-0.026	[-0.20, +0.16]	
chain-pair ipTM	21/31	0.674	[0.52, 0.81]	0.0361	0.0642	-0.026	[-0.20, +0.16]	
ranking score	21/31	0.640	[0.48, 0.79]	0.0877	0.1169	-0.061	[-0.26, +0.14]	
pTM	21/31	0.562	[0.40, 0.72]	0.4512	0.4532	-0.138	[-0.37, +0.10]	
fraction disordered	21/31	0.439	[0.28, 0.60]	0.4532	0.4532	-0.262	[-0.49, -0.01]	
chain-pair PAE min	21/31	0.735	[0.58, 0.87]	0.0043	0.0200	+0.035	[-0.14, +0.22]	
interchain PAE mean	21/31	0.630	[0.47, 0.78]	0.1202	0.1479	-0.071	[-0.26, +0.13]	
interchain PAE p05	21/31	0.685	[0.53, 0.82]	0.0252	0.0576	-0.015	[-0.20, +0.18]	
PAE frac < 10 A	21/31	0.680	[0.54, 0.81]	0.0171	0.0456	-0.020	[-0.19, +0.16]	
PAE frac < 5 A	21/31	0.672	[0.54, 0.80]	0.0154	0.0456	-0.028	[-0.21, +0.16]	
contact prob. sum	21/31	0.798	[0.65, 0.92]	0.0004	0.0064	+0.098	[-0.03, +0.24]	
contact prob. density	21/31	0.665	[0.51, 0.81]	0.0459	0.0734	-0.035	[-0.21, +0.15]	
contact prob. max	21/31	0.730	[0.58, 0.86]	0.0050	0.0200	+0.030	[-0.13, +0.20]	
contacts > 0.5	21/31	0.721	[0.59, 0.85]	0.0026	0.0200	+0.021	[-0.15, +0.21]	
contact particip. ratio	21/31	0.643	[0.49, 0.79]	0.0843	0.1169	-0.058	[-0.19, +0.09]	
mean pLDDT	21/31	0.412	[0.25, 0.59]	0.2891	0.3304	-0.289	[-0.55, -0.01]	

Appendix B continued

By positive subgroup / LLPS-reported positives (15 positive / 31 negative, length alone 0.690, composition 0.794)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	15/31	0.604	[0.44, 0.77]	0.2576	0.3170	-0.189	[-0.35, -0.03]
chain-pair ipTM	15/31	0.604	[0.44, 0.77]	0.2576	0.3170	-0.189	[-0.35, -0.03]
ranking score	15/31	0.567	[0.39, 0.75]	0.4684	0.5353	-0.227	[-0.42, -0.03]
pTM	15/31	0.448	[0.28, 0.63]	0.5763	0.5895	-0.345	[-0.57, -0.10]
fraction disordered	15/31	0.385	[0.22, 0.56]	0.1940	0.3170	-0.409	[-0.65, -0.14]
chain-pair PAE min	15/31	0.667	[0.49, 0.83]	0.0658	0.1754	-0.127	[-0.30, +0.04]
interchain PAE mean	15/31	0.551	[0.38, 0.72]	0.5895	0.5895	-0.243	[-0.42, -0.06]
interchain PAE p05	15/31	0.613	[0.44, 0.78]	0.2162	0.3170	-0.181	[-0.36, -0.00]
PAE frac < 10 Å	15/31	0.619	[0.47, 0.77]	0.1370	0.3131	-0.174	[-0.33, -0.03]
PAE frac < 5 Å	15/31	0.600	[0.46, 0.75]	0.1870	0.3170	-0.194	[-0.35, -0.03]
contact prob. sum	15/31	0.801	[0.62, 0.95]	0.0006	0.0096	+0.008	[-0.11, +0.12]
contact prob. density	15/31	0.606	[0.44, 0.77]	0.2385	0.3170	-0.187	[-0.33, -0.04]
contact prob. max	15/31	0.683	[0.51, 0.84]	0.0442	0.1414	-0.111	[-0.25, +0.03]
contacts > 0.5	15/31	0.669	[0.52, 0.82]	0.0245	0.0980	-0.125	[-0.29, +0.03]
contact particip. ratio	15/31	0.709	[0.54, 0.86]	0.0215	0.0980	-0.085	[-0.23, +0.06]
mean pLDDT	15/31	0.269	[0.12, 0.45]	0.0108	0.0864	-0.525	[-0.78, -0.22]

By positive subgroup / Positives without LLPS report (2 positive / 31 negative, length alone 0.565, composition 0.565)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	2/31	0.661	[0.45, 0.87]	n/a	n/a	+0.097	[-0.13, +0.32]
chain-pair ipTM	2/31	0.661	[0.45, 0.87]	n/a	n/a	+0.097	[-0.13, +0.32]
ranking score	2/31	0.573	[0.18, 0.94]	n/a	n/a	+0.008	[-0.42, +0.42]
pTM	2/31	0.653	[0.42, 0.87]	n/a	n/a	+0.089	[-0.16, +0.35]
fraction disordered	2/31	0.387	[0.13, 0.71]	n/a	n/a	-0.177	[-0.52, +0.21]
chain-pair PAE min	2/31	0.790	[0.55, 1.00]	n/a	n/a	+0.226	[-0.03, +0.48]
interchain PAE mean	2/31	0.597	[0.34, 0.84]	n/a	n/a	+0.032	[-0.23, +0.29]
interchain PAE p05	2/31	0.677	[0.39, 0.94]	n/a	n/a	+0.113	[-0.16, +0.39]
PAE frac < 10 Å	2/31	0.597	[0.31, 0.94]	n/a	n/a	+0.032	[-0.29, +0.39]
PAE frac < 5 Å	2/31	0.621	[0.35, 0.94]	n/a	n/a	+0.056	[-0.24, +0.39]
contact prob. sum	2/31	0.677	[0.26, 1.00]	n/a	n/a	+0.113	[-0.32, +0.55]
contact prob. density	2/31	0.597	[0.29, 0.90]	n/a	n/a	+0.032	[-0.29, +0.35]
contact prob. max	2/31	0.645	[0.32, 0.94]	n/a	n/a	+0.081	[-0.26, +0.42]
contacts > 0.5	2/31	0.694	[0.34, 1.00]	n/a	n/a	+0.129	[-0.26, +0.55]
contact particip. ratio	2/31	0.597	[0.42, 0.77]	n/a	n/a	+0.032	[-0.19, +0.26]
mean pLDDT	2/31	0.468	[0.23, 0.71]	n/a	n/a	-0.097	[-0.39, +0.19]

Length-adjusted / 1. positives vs real negatives (adj.) (17 positive / 31 negative, length alone 0.676, composition 0.767)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	17/31	0.647	[0.48, 0.80]	0.0967	0.1818	-0.120	[-0.28, +0.04]
chain-pair ipTM	17/31	0.647	[0.48, 0.80]	0.0967	0.1818	-0.120	[-0.28, +0.04]
ranking score	17/31	0.620	[0.45, 0.78]	0.1772	0.2577	-0.146	[-0.33, +0.03]
pTM	17/31	0.550	[0.39, 0.71]	0.5793	0.6604	-0.216	[-0.43, -0.00]
fraction disordered	17/31	0.444	[0.28, 0.62]	0.5451	0.6604	-0.323	[-0.55, -0.09]
chain-pair PAE min	17/31	0.784	[0.62, 0.92]	0.0008	0.0088	+0.017	[-0.12, +0.15]
interchain PAE mean	17/31	0.622	[0.46, 0.78]	0.1678	0.2577	-0.144	[-0.30, +0.00]
interchain PAE p05	17/31	0.765	[0.62, 0.89]	0.0021	0.0088	-0.002	[-0.13, +0.13]
PAE frac < 10 Å	17/31	0.545	[0.38, 0.70]	0.6191	0.6604	-0.222	[-0.40, -0.04]
PAE frac < 5 Å	17/31	0.611	[0.45, 0.76]	0.2179	0.2905	-0.156	[-0.35, +0.04]
contact prob. sum	17/31	0.763	[0.59, 0.91]	0.0022	0.0088	-0.004	[-0.16, +0.14]
contact prob. density	17/31	0.765	[0.59, 0.92]	0.0020	0.0088	-0.002	[-0.12, +0.12]
contact prob. max	17/31	0.744	[0.57, 0.89]	0.0054	0.0173	-0.023	[-0.15, +0.09]
contacts > 0.5	17/31	0.712	[0.54, 0.86]	0.0156	0.0416	-0.055	[-0.22, +0.10]
contact particip. ratio	17/31	0.645	[0.49, 0.80]	0.1023	0.1818	-0.121	[-0.28, +0.04]
mean pLDDT	17/31	0.484	[0.31, 0.65]	0.8718	0.8718	-0.283	[-0.53, -0.02]

Appendix B continued

Length-adjusted / 2b. fuzzy positives vs real negatives (adj.) (17 positive / 28 negative, length alone 0.641, composition 0.784)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	17/28	0.712	[0.54, 0.86]	0.0169	0.0338	-0.071	[-0.22, +0.07]
chain-pair ipTM	17/28	0.712	[0.54, 0.86]	0.0169	0.0338	-0.071	[-0.22, +0.07]
ranking score	17/28	0.681	[0.50, 0.84]	0.0436	0.0726	-0.103	[-0.28, +0.08]
pTM	17/28	0.607	[0.43, 0.77]	0.2399	0.2952	-0.176	[-0.39, +0.03]
fraction disordered	17/28	0.466	[0.29, 0.64]	0.7189	0.7189	-0.317	[-0.55, -0.08]
chain-pair PAE min	17/28	0.811	[0.65, 0.94]	0.0002	0.0016	+0.027	[-0.11, +0.16]
interchain PAE mean	17/28	0.685	[0.52, 0.84]	0.0454	0.0726	-0.099	[-0.25, +0.04]
interchain PAE p05	17/28	0.821	[0.68, 0.94]	<0.0001	0.0016	+0.038	[-0.07, +0.15]
PAE frac < 10 Å	17/28	0.601	[0.43, 0.76]	0.2709	0.3096	-0.183	[-0.37, -0.00]
PAE frac < 5 Å	17/28	0.645	[0.48, 0.80]	0.1113	0.1619	-0.139	[-0.34, +0.05]
contact prob. sum	17/28	0.775	[0.60, 0.92]	0.0014	0.0075	-0.008	[-0.17, +0.13]
contact prob. density	17/28	0.775	[0.60, 0.92]	0.0026	0.0083	-0.008	[-0.12, +0.11]
contact prob. max	17/28	0.786	[0.61, 0.93]	0.0019	0.0076	+0.002	[-0.11, +0.11]
contacts > 0.5	17/28	0.763	[0.60, 0.91]	0.0040	0.0107	-0.021	[-0.18, +0.12]
contact particip. ratio	17/28	0.632	[0.46, 0.79]	0.1474	0.1965	-0.151	[-0.32, +0.01]
mean pLDDT	17/28	0.534	[0.35, 0.71]	0.7093	0.7189	-0.250	[-0.50, +0.01]

Length-adjusted / 3. positives vs gold negatives (adj.) (17 positive / 8 negative, length alone 0.985, composition 0.640)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	17/8	0.449	[0.19, 0.72]	0.7118	0.8135	-0.191	[-0.50, +0.09]
chain-pair ipTM	17/8	0.449	[0.19, 0.72]	0.7118	0.8135	-0.191	[-0.50, +0.09]
ranking score	17/8	0.463	[0.20, 0.75]	0.7948	0.8478	-0.176	[-0.51, +0.15]
pTM	17/8	0.346	[0.10, 0.62]	0.2485	0.7384	-0.294	[-0.60, +0.03]
fraction disordered	17/8	0.368	[0.13, 0.62]	0.3231	0.7384	-0.272	[-0.59, +0.05]
chain-pair PAE min	17/8	0.647	[0.43, 0.85]	0.2465	0.7384	+0.007	[-0.20, +0.21]
interchain PAE mean	17/8	0.360	[0.11, 0.64]	0.2717	0.7384	-0.279	[-0.57, +0.00]
interchain PAE p05	17/8	0.507	[0.27, 0.74]	0.9767	0.9767	-0.132	[-0.41, +0.10]
PAE frac < 10 Å	17/8	0.397	[0.12, 0.72]	0.4236	0.8135	-0.243	[-0.59, +0.11]
PAE frac < 5 Å	17/8	0.721	[0.38, 0.99]	0.0856	0.7384	+0.081	[-0.32, +0.43]
contact prob. sum	17/8	0.632	[0.40, 0.86]	0.3138	0.7384	-0.007	[-0.22, +0.21]
contact prob. density	17/8	0.632	[0.40, 0.85]	0.3213	0.7384	-0.007	[-0.24, +0.21]
contact prob. max	17/8	0.551	[0.32, 0.78]	0.6899	0.8135	-0.088	[-0.35, +0.15]
contacts > 0.5	17/8	0.551	[0.29, 0.80]	0.6901	0.8135	-0.088	[-0.41, +0.21]
contact particip. ratio	17/8	0.559	[0.32, 0.82]	0.6599	0.8135	-0.081	[-0.40, +0.23]
mean pLDDT	17/8	0.441	[0.17, 0.73]	0.6707	0.8135	-0.199	[-0.55, +0.17]

Length-adjusted / Homotypic only (adj.) (12 positive / 11 negative, length alone 0.795, composition 0.902)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	12/11	0.735	[0.50, 0.92]	0.0595	0.0952	-0.167	[-0.39, +0.04]
chain-pair ipTM	12/11	0.735	[0.50, 0.92]	0.0595	0.0952	-0.167	[-0.39, +0.04]
ranking score	12/11	0.765	[0.55, 0.95]	0.0311	0.0622	-0.136	[-0.38, +0.10]
pTM	12/11	0.545	[0.29, 0.78]	0.7368	0.8302	-0.356	[-0.60, -0.14]
fraction disordered	12/11	0.470	[0.22, 0.71]	0.8302	0.8302	-0.432	[-0.68, -0.19]
chain-pair PAE min	12/11	0.962	[0.87, 1.00]	0.0002	0.0024	+0.061	[-0.04, +0.20]
interchain PAE mean	12/11	0.697	[0.46, 0.90]	0.1126	0.1638	-0.205	[-0.43, +0.00]
interchain PAE p05	12/11	0.909	[0.77, 1.00]	0.0007	0.0028	+0.008	[-0.13, +0.14]
PAE frac < 10 Å	12/11	0.644	[0.39, 0.87]	0.2560	0.3413	-0.258	[-0.52, -0.01]
PAE frac < 5 Å	12/11	0.795	[0.58, 0.96]	0.0151	0.0345	-0.106	[-0.33, +0.12]
contact prob. sum	12/11	0.932	[0.80, 1.00]	0.0003	0.0024	+0.030	[-0.12, +0.20]
contact prob. density	12/11	0.879	[0.70, 1.00]	0.0012	0.0038	-0.023	[-0.23, +0.16]
contact prob. max	12/11	0.917	[0.78, 1.00]	0.0005	0.0027	+0.015	[-0.11, +0.15]
contacts > 0.5	12/11	0.848	[0.67, 0.98]	0.0036	0.0096	-0.053	[-0.22, +0.11]
contact particip. ratio	12/11	0.470	[0.23, 0.71]	0.8301	0.8302	-0.432	[-0.73, -0.12]
mean pLDDT	12/11	0.614	[0.36, 0.83]	0.3736	0.4598	-0.288	[-0.55, -0.04]

Appendix B continued

Length-adjusted / Heterotypic only (adj.) (5 positive / 20 negative, length alone 0.530, composition 0.500)

metric	n	AUC	95% CI	perm p	BH q	delta	delta 95% CI
ipTM	5/20	0.390	[0.14, 0.66]	0.4953	0.9014	-0.110	[-0.45, +0.21]
chain-pair ipTM	5/20	0.390	[0.14, 0.66]	0.4953	0.9014	-0.110	[-0.45, +0.21]
ranking score	5/20	0.380	[0.16, 0.63]	0.4546	0.9014	-0.120	[-0.45, +0.23]
pTM	5/20	0.570	[0.33, 0.80]	0.6760	0.9014	+0.070	[-0.21, +0.34]
fraction disordered	5/20	0.570	[0.27, 0.85]	0.6746	0.9014	+0.070	[-0.26, +0.41]
chain-pair PAE min	5/20	0.460	[0.20, 0.73]	0.8237	0.9193	-0.040	[-0.37, +0.31]
interchain PAE mean	5/20	0.390	[0.14, 0.66]	0.4935	0.9014	-0.110	[-0.45, +0.21]
interchain PAE p05	5/20	0.540	[0.28, 0.80]	0.7964	0.9193	+0.040	[-0.29, +0.35]
PAE frac < 10 Å	5/20	0.380	[0.13, 0.66]	0.4504	0.9014	-0.120	[-0.46, +0.20]
PAE frac < 5 Å	5/20	0.420	[0.16, 0.70]	0.5998	0.9014	-0.080	[-0.43, +0.26]
contact prob. sum	5/20	0.400	[0.16, 0.67]	0.5321	0.9014	-0.100	[-0.41, +0.20]
contact prob. density	5/20	0.380	[0.10, 0.70]	0.4524	0.9014	-0.120	[-0.49, +0.25]
contact prob. max	5/20	0.380	[0.14, 0.64]	0.4505	0.9014	-0.120	[-0.45, +0.19]
contacts > 0.5	5/20	0.380	[0.13, 0.66]	0.4508	0.9014	-0.120	[-0.46, +0.20]
contact particip. ratio	5/20	0.520	[0.27, 0.76]	0.9193	0.9193	+0.020	[-0.26, +0.30]
mean pLDDT	5/20	0.530	[0.25, 0.80]	0.8735	0.9193	+0.030	[-0.31, +0.37]